

CHAPTER 1

INTRODUCTION AND DESIGN OF THE STUDY

1.1 Introduction

The FMCG (Fast Moving Consumer Goods) industry includes products that are used daily by consumers. Beverages such as fruit juices and flavoured drinks are an important part of this industry. The demand for juice products changes based on seasons, weather conditions, customer taste and market trends. For example, juice sales are usually higher during summer months.

In recent years many companies have started using data and technology to improve their planning and decision-making. Data Science helps in studying past sales data to understand patterns and trends. Time-series forecasting and simple machine learning methods can be used to predict future sales based on historical data. By using such methods, companies can plan their production better, reduce wastage, and manage inventory in a more effective way.

This project focuses on developing a simple sales and demand forecasting framework for juice products using data science methods. Due to data confidentiality, publicly available FMCG and retail sales data is used in this study to demonstrate how forecasting methods can be applied in a juice manufacturing business context.

1.2 Statement of the Problem

Juice manufacturing companies often face difficulty in predicting future sales accurately. Sales may change due to seasonal factors, changes in customer demand and market conditions. When production is planned without proper demand estimation it can result in overproduction and wastage of perishable products or shortage of products in the market.

There is a need for a data-based approach to predict future sales more accurately. A simple forecasting framework using data science methods can help in better production planning and inventory management in the beverage industry.

1.3 Period of Study

The period of study is taking place last five years datasets from the year **2021 to 2025**.

1.4 Need for the Study

There is a growing need for businesses to use data for better decision-making. In the beverage industry, especially for juice products, demand changes frequently and is affected by seasons and customer preferences. Without proper forecasting, companies may face losses due to wastage or missed sales opportunities. Therefore, a simple and practical forecasting approach is needed to support planning activities.

At the same time, many students from commerce backgrounds aim to build careers in data related fields such as Data Science. There is a need to apply data science methods to real business problems in order to gain practical skills. This study helps in understanding how data analysis and forecasting can be applied to a real-world business context related to juice products.

1.5 Tools used for the project

Programming and analysis:

- Python
- Jupyter Notebook

Python Libraries:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn (sklearn)

Data source and Management:

- Microsoft Excel

1.6 Objectives of the Study

The main objectives of this study are:

- To analyze the sales performance of juice products
- To examine the factors influencing sales and demand
- To identify seasonal patterns and demand variations
- To forecast future sales and demand

1.7 Scope of the Study

This study focuses on sales and demand forecasting for juice products using publicly available FMCG and retail sales data. The forecasting framework developed in this project is general in nature and is not based on the actual sales data of any specific company. The company Podaran Foods India Private Limited is referred to as a case study context based on internship experience to explain how the proposed framework can be applied in a real juice manufacturing business. The forecasts generated in this study are meant to demonstrate the method and do not represent the actual future sales of the company.

1.8 Significance of the Study

This study is useful in showing how data science methods can be applied to predict sales and support planning in the beverage industry. The proposed forecasting framework can help businesses understand demand patterns, plan production and reduce wastage of perishable products. From an academic point of view, the study helps students from non-technical backgrounds understand how data science tools can be applied to solve real business problems.

1.9 Limitations of the Study

The limitations of the study are as follows:

1. The study is based on secondary or simulated sales data and does not use actual company records due to confidentiality constraints.
2. The dataset used may not fully represent real-time business conditions, and therefore the results may differ from actual market performance.
3. The study applies basic forecasting techniques such as moving average and trend analysis, which may not capture complex patterns compared to advanced models.
4. The accuracy of the forecasting results depends on the quality, completeness, and consistency of the available data.
5. External factors such as sudden market changes, competitor strategies, weather conditions, and promotional activities are not fully incorporated into the analysis.
6. The study assumes that past trends will continue in the future, which may not always be true in dynamic market conditions.

1.10 Chapter Scheme

This report is organized into five chapters.

Chapter 1 - Introduction and Design of the Study

Chapter 2 - Review of Literature

Chapter 3 - Company Profile

Chapter 4 - Data Analysis and Interpretation

Chapter 5 - Findings, Suggestions and Conclusion

CHAPTER 2

REVIEW OF LITERATURE

2.1 Introduction

Sales forecasting is an important part of planning in manufacturing and retail businesses. Many researchers have studied different methods to predict sales using past data. With the growth of data tools and computer systems, new methods have been developed to improve the accuracy of sales prediction. Studies in the area of demand forecasting show that data-based methods help businesses understand sales patterns, seasonal changes, and customer demand more clearly. This chapter reviews earlier studies related to sales forecasting, time-series analysis, and the use of data science methods in retail and FMCG sectors, which are relevant to juice product sales.

2.2 Literature review

Hyndman and Athanasopoulos (2018)¹ discussed various forecasting techniques used in time-series analysis. The study focused on methods such as moving averages, trend analysis, and seasonal models for predicting future values. They emphasized that historical data plays a crucial role in identifying patterns and trends in sales data. The authors also highlighted that simple forecasting techniques can provide reliable and interpretable results in practical situations. The study concluded that time-series methods are highly useful for short-term demand forecasting in business environments.

Box, Jenkins, Reinsel, and Ljung (2015)² conducted an in-depth study on time-series analysis and forecasting techniques. The research focused on statistical models used to analyze patterns in time-based data. They explained how past data can be used to predict future outcomes through structured modeling approaches. The study emphasized the importance of identifying trends, seasonality, and randomness in data. It concluded that time-series models are effective tools for forecasting sales and supporting business decision-making.

Pavlyshenko (2019)³ analyzed the application of machine learning models in sales forecasting. The study explored how advanced algorithms can capture complex relationships in time-series data. It highlighted that machine learning techniques improve prediction accuracy compared to traditional methods. However, the study also noted that these models require large

datasets and technical expertise. The research concluded that while advanced models are powerful, simpler methods remain useful for practical applications.

Fildes, Ma, and Kolassa (2019)⁴ examined forecasting practices in the retail sector. The study focused on improving forecasting accuracy to enhance business performance. It highlighted that accurate demand forecasting reduces uncertainty and improves operational efficiency. The research emphasized the role of data-driven methods in understanding customer demand patterns. The study concluded that effective forecasting is essential for inventory planning and sales management.

Chopra and Meindl (2019)⁵ discussed the importance of demand forecasting in supply chain management. The study highlighted how forecasting helps in planning production, distribution, and inventory. It emphasized that accurate forecasting reduces costs and improves customer satisfaction. The authors explained that forecasting plays a critical role in aligning supply with demand. The study concluded that demand forecasting is a key component in achieving efficient supply chain performance.

Géron (2019)⁶ explored machine learning techniques used for predictive analysis. The study explained how algorithms can be applied to forecast future trends based on historical data. It highlighted that machine learning models can handle large and complex datasets effectively. However, the study also pointed out that these models require proper training and validation. The research concluded that machine learning enhances forecasting accuracy but requires technical expertise.

McKinney (2018)⁷ focused on the use of Python for data analysis and processing. The study explained how tools like Pandas help in cleaning and organizing datasets. It highlighted that data preprocessing is an important step before analysis. The research also emphasized the role of programming in handling large datasets efficiently. The study concluded that Python is a powerful tool for data analysis and supports effective decision-making.

VanderPlas (2016)⁸ discussed data science techniques and their applications in data analysis. The study highlighted the importance of visualization tools in understanding patterns and trends. It explained how graphical methods help in interpreting complex data easily. The research

emphasized that visualization improves the clarity of analysis. The study concluded that data visualization is essential for effective communication of results.

Pedregosa et al. (2011)⁹ introduced the Scikit-learn library for machine learning applications. The study explained how this library supports various algorithms for data analysis and prediction. It highlighted the flexibility and efficiency of Scikit-learn in handling different datasets. The research emphasized its usefulness in implementing predictive models. The study concluded that Scikit-learn is an important tool for machine learning and data analysis.

Provost and Fawcett (2013)¹⁰ examined the role of data science in business decision-making. The study highlighted how data-driven approaches improve business performance. It emphasized the importance of analyzing data to extract meaningful insights. The research explained how predictive analytics supports forecasting and planning. The study concluded that data science plays a crucial role in modern business environments.

Hanke and Wichern (2014)¹¹ studied various business forecasting techniques used for predicting future trends. The study focused on quantitative methods such as moving averages and trend analysis. It emphasized that these methods are useful for short-term forecasting in business environments. The authors highlighted that accurate forecasting helps in better planning and decision-making. The study concluded that simple forecasting techniques are effective and widely used in practical applications.

Makridakis, Spiliotis, and Assimakopoulos (2018)¹² analyzed different statistical and machine learning forecasting methods. The study compared the performance of various forecasting models in real-world scenarios. It highlighted that simple statistical methods often perform as well as complex models. The research emphasized the importance of selecting appropriate models based on data characteristics. The study concluded that simplicity and accuracy are key factors in effective forecasting.

Makridakis, Hyndman, and Petropoulos (2020)¹³ examined forecasting methods through the M4 competition. The study evaluated multiple forecasting models across different datasets. It highlighted that combining different forecasting approaches improves prediction accuracy. The research emphasized the importance of model selection and evaluation. The study concluded that hybrid approaches can enhance forecasting performance.

Taylor and Letham (2018)¹⁴ discussed scalable forecasting techniques for large datasets. The study introduced methods that can handle time-series data efficiently. It emphasized the importance of automated forecasting systems in business environments. The research highlighted that scalable models improve forecasting accuracy and efficiency. The study concluded that advanced forecasting methods are useful for large-scale data analysis.

Zhang (2003)¹⁵ studied hybrid forecasting models combining ARIMA and neural networks. The research focused on improving forecasting accuracy by combining statistical and machine learning methods. It highlighted that hybrid models capture both linear and non-linear patterns in data. The study emphasized the advantages of combining different approaches. The study concluded that hybrid models provide better forecasting results than individual models.

Montgomery, Jennings, and Kulahci (2015)¹⁶ explained time-series forecasting techniques in detail. The study focused on identifying trends, seasonality, and patterns in data. It emphasized the importance of analyzing historical data for forecasting purposes. The research highlighted that proper modeling improves prediction accuracy. The study concluded that time-series analysis is essential for business forecasting.

Shumway and Stoffer (2017)¹⁷ discussed applications of time-series analysis in forecasting. The study explained how data collected over time can be used to predict future outcomes. It emphasized the importance of understanding temporal patterns in data. The research highlighted that forecasting models help in identifying trends and variations. The study concluded that time-series analysis is a valuable tool for decision-making.

Goodwin (2010)¹⁸ studied exponential smoothing techniques used in forecasting. The research focused on methods that capture trends and seasonal patterns in data. It highlighted that smoothing techniques reduce fluctuations in data. The study emphasized that these methods are useful for short-term forecasting. The study concluded that exponential smoothing improves the accuracy of predictions.

Carbonneau, Laframboise, and Vahidov (2008)¹⁹ applied machine learning techniques in supply chain demand forecasting. The study explored how predictive models improve forecasting accuracy. It highlighted that machine learning models can handle complex data

patterns. The research emphasized the role of technology in improving demand prediction. The study concluded that advanced techniques enhance forecasting performance.

Waller and Fawcett (2013)²⁰ examined the role of data science and predictive analytics in supply chain management. The study highlighted that data-driven decision-making improves business performance. It emphasized the importance of big data in forecasting and planning. The research showed that analytics helps in understanding demand patterns. The study concluded that predictive analytics enhances operational efficiency.

Mentzer and Moon (2005)²¹ discussed sales forecasting management in business organizations. The study emphasized the importance of forecasting in improving business planning. It highlighted that accurate forecasting reduces uncertainty and improves decision-making. The research explained how forecasting supports inventory and production management. The study concluded that sales forecasting is essential for business success.

Lapide (2006)²² examined the difference between demand forecasting and sales forecasting. The study highlighted that both concepts are important in business planning. It emphasized that demand forecasting focuses on customer needs, while sales forecasting focuses on company performance. The research explained the relationship between the two forecasting methods. The study concluded that both approaches are necessary for effective planning.

Bandara, Bergmeir, and Smyl (2020)²³ studied forecasting techniques using neural networks. The research focused on handling large-scale time-series datasets. It highlighted that deep learning models can capture complex patterns in data. The study emphasized the importance of advanced models in modern forecasting. The study concluded that neural networks improve forecasting accuracy in large datasets.

Kaggle (2024)²⁴ provides publicly available datasets for research and analysis. The study platform offers various datasets related to retail and FMCG sales. It helps researchers analyze real-world data for forecasting purposes. The platform supports data-driven learning and experimentation. The study concluded that such datasets are valuable for practical forecasting studies.

UCI Machine Learning Repository (2024)²⁵ offers a wide range of datasets for academic research. The repository includes datasets related to retail and business analysis. It supports data analysis and forecasting studies using real-world data. The platform is widely used by researchers and students for experimentation. The study concluded that publicly available datasets enhance data-driven research.

2.3 References

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CHAPTER 3

COMPANY PROFILE

3.1 Introduction of the Company

Podaran Foods India Private Limited is a Tamil Nadu-based private limited company engaged in the manufacture of fruit-based beverages, specializing in the production of hygienically processed and packaged fruit juices. Located in Melapalayam, Nakakadayur, the company has earned a reputation in the regional market for its quality-focused production and cost-effective distribution of natural juice products. The company was founded with a clear mission to bridge the gap between traditional fruit consumption and modern beverage convenience. In line with this, Podaran Foods has designed its operations around a semiautomated production line that emphasizes efficiency, hygiene and consistency.

The factory is equipped with specialized equipment for fruit washing, pulp extraction, filtration, pasteurization, blending, bottling, sealing, labelling and final packaging. This systematic process ensures that each batch meets the company's internal quality benchmarks and complies with FSSAI (Food Safety and Standards Authority of India) regulations. Podaran Foods also emphasizes worker safety, environmental cleanliness and resource efficiency, aligning itself with responsible manufacturing practices. The factory operates on a two-shift schedule, enabling continuous production while ensuring effective supervision and maintenance. The management fosters a collaborative work environment, where cross-functional coordination between production, packaging and logistics teams ensures timely and safe delivery of finished goods.

By continually investing in its people, processes, and product development, Podaran Foods India Private Limited aspires to become a leading player in the Indian juice market. It remains committed to contributing to public health, economic development, and responsible manufacturing through every bottle it produces. By combining local sourcing of raw materials, a dedicated team and a strong commitment to food safety, Podaran Foods India Private Limited has positioned itself as a promising player in South India's juice manufacturing sector. The organization continues to grow steadily by expanding its product offerings and strengthening its presence in both urban and rural retail markets.

3.2 Company Profile

PARTICULARS	DETAILS
Name of the company	Podaran Foods India Private Limited
Address	461/1, Melapalayam, Chitampalayam, Nathakadayur via, Kangayam Taluk, Tirupur District, Tamil Nadu – 638108
Head office	Melapalayam, Nakakadayur, Tirupur District, Tamil Nadu
Managing director	Mr. P. Suresh Kumar
Business type	Private Limited Company
Website	https://www.podaranfoods.com/
Accredited	FSSAI Approved, ISO 22000 (Food Safety)
Industry	Food & Beverage – Fruit Juices & Soft Drink Manufacturing
GST no.	33AAHCP3183F1ZJ
Establishment year	1969
CIN no.	U15400TZ2013PTC019718
Turnover	₹347 crore
Email	md@podaranfoods.com
Logo	

3.3 Vision and Mission

Vision

“To become a leading food and beverage manufacturer in South India, offering safe, affordable and hygienically processed products that satisfy customer needs and taste preferences.”

Mission

- To consistently improve food safety and manufacturing efficiency.
- To invest in semi-automated technology and a trained workforce.
- To deliver products that meet both local taste and quality expectations.
- To follow environmentally conscious and sustainable practices in operations.

3.4 Core Values

Quality: The company is committed to maintaining high quality standards through regular testing and strict adherence to food safety protocols. Every product is checked for taste, hygiene, and consistency.

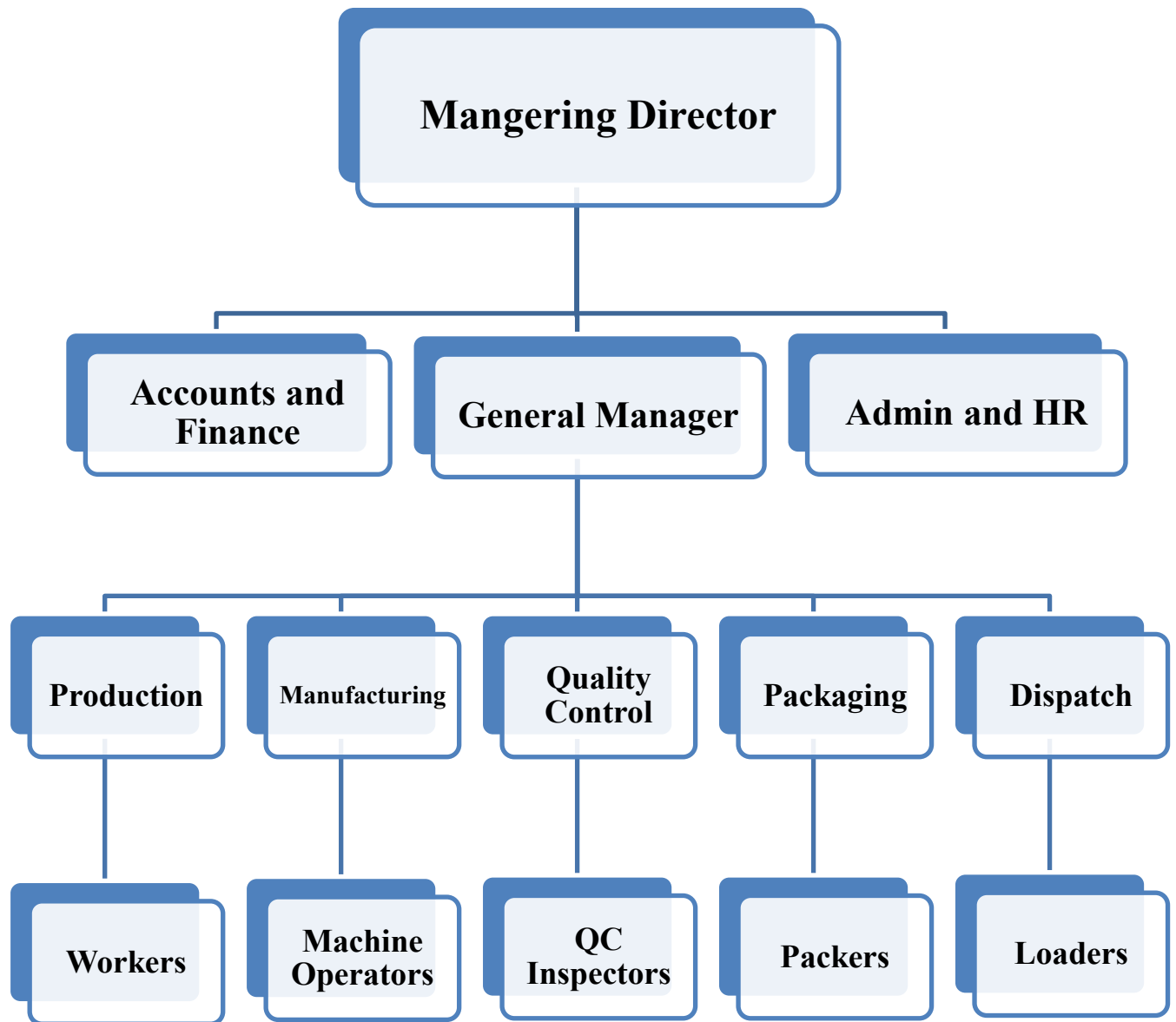
Customer Satisfaction: Customer needs and preferences are prioritized. Products are made affordable and tailored to regional tastes, with a focus on timely delivery and appealing packaging.

Hygiene and Safety: Strict hygiene practices are followed in all departments. Employees use protective gear and production areas are cleaned regularly to ensure safe food handling.

Teamwork and Collaboration: Podaran believes that successful production and delivery of goods require effective teamwork. Departments such as Production, Manufacturing, Quality Control, Packaging, and Logistics work together in coordination.

Innovation: While Podaran retains its traditional values and taste legacy, it also embraces innovation. The company regularly upgrades its machinery, packaging systems, and product offerings to remain competitive

3.5 Organisational Chart



3.6 Products of the Company

3.6.1 Carbonated Soft Drinks



Carbonated soft drinks are one of the major products manufactured by Podaran Foods India Pvt. Ltd. These beverages are made by blending purified water with flavoured syrup and carbon dioxide gas (CO₂) under controlled pressure. The resulting drink is fizzy, refreshing, and available in various regional flavours to suit consumer preferences.

Flavours Produced:

- Cola
- Orange
- Lemon
- Mango
- Paneer soda
- Jeera masala
- Power cola

3.6.2 Fruit Juice



Podaran Foods India Pvt. Ltd. manufactures a variety of non-carbonated fruit juices made using real fruit pulp or concentrates. These juices are sweetened, pasteurized, and packed under hygienic conditions to ensure taste, safety, and shelf life. They are especially popular among health-conscious consumers, schoolchildren, and families. The juices are available in different flavours and are filled into PET bottles ranging from 200 ml to 1 litre.

Types of Fruit Juices Produced:

- Mango
- Apple
- Guava
- Mixed fruit

3.6.3 Flavoured Drinks



Flavoured drinks are a popular product segment manufactured by Podaran Foods India Pvt. Ltd. These beverages are especially appealing to regional markets that enjoy traditional and health-oriented soft drinks. The company focuses on providing refreshing options that not only quench thirst but also offer functional benefits such as digestion and cooling.

The flavoured drinks are typically mildly carbonated, hygienically processed, and attractively packaged. These drinks are in high demand during summer seasons and festive occasions.

Types of Flavoured Drinks Produced:

- Paneer soda
- Jeera masala
- Lemonade

3.6.4 Milk-Based Products



Podaran Foods India Pvt. Ltd. also operates a dairy beverage division that produces a range of milk-based drinks. These beverages are nutritious and cater to the growing demand for healthy, ready-to-drink dairy options. Milk-based beverages are produced in flavoured variants and are enriched with milk solids, sugar, and permitted natural or nature-identical flavours. They are especially consumed by children, students, and people seeking an energy drink during work or travel. All milk drinks are prepared under pasteurized conditions and stored in chilled environments to maintain safety and quality.

Types of Milk-Based Beverages:

- Badam milk
- Pista milk
- Chocolate milk
- Strawberry milk

3.7 Departments of the Company

Podaran Foods India Pvt. Ltd., located in Nathakadaiyur, Tamil Nadu, is engaged in the manufacturing and distribution of beverages such as carbonated soft drinks, fruit juices, flavoured water and milk-based drinks. The company operates through a structured departmental workflow to ensure smooth and hygienic operations from raw material intake to final dispatch.

During my internship, I had the opportunity to focus on the core operational departments involved in real-time beverage processing. These departments are interlinked and work systematically to convert raw materials into packaged, ready-to-sell beverages.

The company has the following departments functioning under one integrated manufacturing facility:

1. Raw Material Storage & Inventory
2. Production Department
3. Manufacturing Department
4. Quality Control Department
5. Packaging Department
6. Dispatch & Logistics Department
7. Maintenance Department
8. Procurement / Purchase Department
9. Accounts & Billing Department
10. Administration / HR Department

3.7.1 Focused Departments

The following departments were observed and focused:

- Production Department
- Manufacturing Department
- Quality Control Department
- Packaging Department

These departments form the backbone of Podaran's beverage processing unit and are crucial for ensuring quality, hygiene, and efficiency in day-to-day operations.

3.7.2 Production Department

The Production Department is where the journey of a Podaran drink begins. It involves the preparation of raw materials such as sugar, fruit pulp, RO water, flavouring essence, and preservatives. These are then mixed to form the base juice or soft drink syrup. The department plays a critical role as it determines the taste, consistency, and stability of the beverage before it goes into bottling. The Production Department is where the journey of a Podaran drink begins. It involves the preparation of raw materials such as sugar, fruit pulp, RO water, flavouring essence, and preservatives. These are then mixed to form the base juice or soft drink syrup. The department plays a critical role as it determines the taste, consistency, and stability of the beverage before it goes into bottling. (e.g., Paneer Soda, Cola, Lemon, etc.). For thick juices like Guava or Mango, real pulp is used, which is procured from suppliers in Bangalore and other regions.

At Podaran, the production area is neat, temperature-controlled, and organized, with designated storage areas for raw materials. The production team follows SOPs (Standard Operating Procedures) and works with a combination of manual handling and semi-automated mixing systems. Hygiene is strictly maintained here. All workers wear gloves, head caps, and coats. The area is cleaned regularly and standard operating procedures (SOPs) are followed for every batch.

Objectives of the Production Department:

- To prepare safe and consistent beverage syrup
- To maintain hygiene during the raw material mixing process
- To ensure accurate ingredient measurement and filtration
- To store prepared syrup in clean, temperature-controlled tanks



Storage Room

Daily Process Flow:

1.Receipt of Raw Materials

Raw materials like fruit pulp (mango, guava, etc.), sugar sacks, essence bottles, and citric acid arrive and are checked for quality.

2. Storage & Sorting

Materials are stored in a clean warehouse. Labels and usage instructions are marked. Workers wear caps and gloves during handling.

3.Mixing Process

A large stainless-steel mixing tank (over 12 feet high) is used. Workers climb built-in stairs to pour in sugar and essence are marked. Workers wear caps and gloves during handling.

4.Batch Preparation Time

Each product takes about 10–12 hours to prepare and is usually made for 2 continuous days per batch. For example, if Paneer Soda is prepared today and tomorrow, Cola or Guava might be prepared the next day.

5. Filtration

The mixed juice passes through an industrial filtration unit to remove impurities and ensure it meets clarity standards.

6.Cooling & Transfer

The filtered juice is transferred to chilled storage tanks that maintain stable temperatures and prevent early spoilage.



Mixing Process

Machinery & Equipment used:

- Stainless Steel Syrup Mixers (12+ feet tall)
- Manual weighing scale for sugar/essence
- Filtration System with micron filters
- Cooled Storage Tanks (pressurized, insulated)

Safety, Hygiene & Team Practices

All staff in the production unit wear:

- Head caps
- Gloves
- Apron-style coats (These prevents contamination)
- The production area is mopped regularly.

3.7.3 Manufacturing Department



Moulding

Once the beverage syrup is prepared in the production department, it moves to the Manufacturing Department, which is responsible for transforming the syrup into a finished product. This includes bottle blowing, cooling, juice filling, and initial conveyor checks.

Podaran Foods has semi-automated systems in this unit, and it was one of the most exciting departments I visited because I saw bottles being “born” from tiny preforms, then cooled and filled with fresh juice. The manufacturing department is fast-paced and focused on precision, hygiene, and consistency.

Objectives of the Manufacturing Department

- To blow PET preforms into bottles of required sizes (typically 200 ml)
- To cool down the newly formed bottles before filling
- To fill each bottle with the correct beverage according to batch schedule
- To monitor machines for faults, fill-level issues, or delays
- To maintain hygienic practices throughout the production cycle



Bottle Manufacturing

Process Flow:

1. Bottle Blowing Process:

There are thousands of tiny, thick plastic pieces (called preforms) about the size of a pinky finger. These are placed into a blowing machine which heats the preforms and stretches them into 200 ml PET bottles using compressed air and moulds.

2. Bottle Cooling System:

Newly blown bottles are hot and must be cooled before filling. They are sent through a cooling tunnel, and this through a glass panel. If bottles are not cooled properly, the juice's flavour and shelf life can be affected.

3. Juice Filling Process:

After cooling, bottles are moved by conveyor belts to the filling area. Here, juice from the chilled tanks is filled into the bottles via automated filling heads.

4. Belt Monitoring & Staff Duties:

Employees are stationed near the machines to check for any bottle jams, spills, or errors. If a machine stalls or malfunctions (like a stuck bottle), a beeping alert sound. This happen once and a staff member quickly fixed it.

Machines & Technology used:

- Blow Moulding Machine (for PET bottle formation)
- Compressed Air Compressor (for bottle expansion)
- Bottle Cooling Tunnel
- Juice Filling Line (semi-automatic, high-speed)
- Conveyor Belt Monitoring Sensors

Safety, Cleanliness & Standards:

- The area is dry-mopped frequently to avoid spills
- Workers use non-slip shoes, gloves, and head caps
- Machines are cleaned after each juice batch, especially nozzles and belts
- Temperature and bottle GSM are monitored regularly

3.7.4 Quality Control Department



Quality Check

The Quality Control (QC) Department is one of the most crucial departments in any food and beverage manufacturing company — and Podaran Foods India Pvt. Ltd. is no exception. This department ensures that every drink manufactured is safe, hygienic, and consistent in taste, volume, and appearance before it reaches the consumer.

There was multiple quality control measures integrated directly on the factory floor, including filtration checks, sensor-based volume validation, and manual inspections for cleanliness and accuracy. This shows that quality at Podaran is not just a step but a continuous process.

Objectives of the Quality Control Department:

- To ensure the final product is safe for human consumption
- To verify accurate volume and consistency of each bottle
- To check hygiene and sanitation during production
- To inspect bottles for physical defects or errors
- To validate that labelling and dates are accurate and legible

Quality Control Process:

1. Filtration as First Quality Gate:

The first major quality control checkpoint happens in the Production Department itself - via a filtration unit. This system removes any undissolved particles, clumps, or contaminants that may remain after mixing.

2. Bottle Volume Verification:

After filling, each bottle is scanned by a sensor-based system that checks if it contains the correct amount - exactly 200 ml for standard-size products.

3. Manual Observations by Staff:

Near every critical point (filling, capping, labelling), workers continuously watch:

- Loose or missing caps
- Spills on the bottle body
- Discoloration or turbidity in the liquid
- Any signs of leakage

4. Label and Print Check:

At the labelling station, the MRP, batch number, manufacturing date, and use-by date are printed using a high-speed machine. However, final legibility checks are done by both sensors and human workers to ensure compliance with labelling standards.

Machines and Tools involved in Quality Control:

- Industrial Filtration Machine (micron level)
- Volume Sensor System
- Label Printing Gun (for date, MRP)
- Defect Rejection Arms (mechanical pushers)
- Manual Inspection Lights at QC stations

3.7.5 Packaging Department



Labelling

The Packaging Department at Podaran Foods India Pvt. Ltd. is where the final transformation of the product takes place — from a plain filled bottle to a branded, properly labelled and market-ready product. This department ensures that each beverage is not only protected for storage and transport but also appeals to consumers through professional presentation.

Packaging is both a technical and visual process. It combines machinery (like shrink wrappers, printers, conveyors) and manual supervision to deliver accuracy, hygiene, and branding compliance.

Objectives of the Packaging Department:

- To label each bottle with correct branding and product details
- To print batch number, manufacturing date, expiry, and MRP
- To detect and remove defective or damaged bottles
- To group bottles into sets and wrap them for safe handling
- To prepare the finished product for warehouse storage and dispatch

Packaging Process:

1. Bottle Transfer from Filling to Packaging Line

After the bottles are filled and capped, they are transferred via conveyor belts into the packaging zone. Separate belts are used for different product types to avoid confusion or misplacement.

2. Labelling Machines in Action

Each bottle is passed through a high-speed labelling machine that applies branded labels (e.g., for Paneer Soda, Lemon Soda, Cola). These labels contain:

- Brand name and logo
- Ingredients list
- Product volume (e.g., 200 ml)
- Manufacturing company details

3. MRP and Date Printing

There is a gun-like printing device that stamps:

- Manufacturing Date
- Expiry/Use by Date
- Maximum Retail Price (MRP). This happens automatically as bottles pass under the printer. Accuracy is checked both by sensor and manual visual check.

4. Defect Detection & Rejection

After labelling and printing, a quality sensor checks for:

- Improperly stuck labels
- Blurred printing
- Scratched or dented bottles

5. Shrink Wrapping into Sets

Bottles are grouped into sets of 12 using a wrapping machine. A thin plastic film wraps these sets tightly and evenly, preparing them for safe stacking and transport.

6. Pallet Stacking and Warehouse Transfer

Wrapped sets are then arranged on pallets using a semi-automated stacker or manually by workers. Pallets are sorted based on product type and moved to the warehouse where they await loading into delivery trucks.



Stacking

Machines and Tools Involved:

- Conveyor Belt System
- Automatic Labelling Machine
- Label Gun
- Defect Sensor & Rejection Arm
- Shrink Wrapping Machine
- Pallet Stacker (semi-robotic)

CHAPTER 4

DATA ANALYSIS AND INTERPRETATION

4.1 Introduction

This chapter presents the analysis of sales and demand data of juice products using various data science techniques. The data has been analyzed using statistical tools and visualization methods to understand trends, patterns and relationships. The objective of this chapter is to interpret the data and provide meaningful insights for forecasting future demand.

4.2 Tools and Technologies used

The following tools and technologies were used for data analysis and forecasting:

- Python – Used for performing data analysis and forecasting.
- Pandas – Used for data cleaning, manipulation and preprocessing.
- Matplotlib – Used for creating graphical visualizations.
- Seaborn – Used for advanced visualization such as heatmaps and regression analysis.
- Jupyter Notebook – Used as the platform for coding and analysis.
- CSV Files – Used as the data source for storing sales data.

4.3 Sample Code Implementation

- **Importing Libraries**

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

- **Loading and Combining Data**

```
df1 = pd.read_csv("podaran_21.csv")
```

```
df2 = pd.read_csv("podaran_22.csv")
```

```
df3 = pd.read_csv("podaran_23.csv")
```

```
df4 = pd.read_csv("podaran_24.csv")
```

```
df5 = pd.read_csv("podaran_25.csv")
df = pd.concat([df1, df2, df3, df4, df5], ignore_index=True)
```

- **Data Preprocessing**

```
df['Date'] = pd.to_datetime(df['Date'])
df = df.sort_values('Date')
df_copy = df.copy()
df_copy.set_index('Date', inplace=True)
```

- **Basic Data Analysis**

```
print(df_copy.info())
print(df_copy.describe())
```

4.4 Data Description

The dataset used for this study consists of sales data from the year 2021 to 2025. It includes various variables such as product name, bottle size, region, production quantity, sales quantity, price, revenue, marketing spend and customer rating.

The dataset is structured to reflect real-world business conditions and is suitable for analyzing demand patterns and forecasting sales.

4.5 Exploratory Data Analysis (EDA)

Table no 4.5.1 (2021 – 2023 Datasets)

Date	Sales_Qty	Date	Sales_Qty	Date	Sales_Qty
02-11-2021	333	14-11-2022	370	19-10-2023	450
05-11-2021	495	16-02-2022	547	30-10-2023	220
10-03-2021	267	27-01-2022	278	06-02-2023	110
03-09-2021	246	02-01-2022	777	23-04-2023	393
19-01-2021	262	31-05-2022	669	20-06-2023	108
28-06-2021	199	30-09-2022	579	12-06-2023	481
05-05-2021	675	21-08-2022	818	03-03-2023	801
23-12-2021	115	12-08-2022	564	30-05-2023	240
21-12-2021	282	24-01-2022	342	30-12-2023	168
28-05-2021	516	17-07-2022	312	27-07-2023	233
14-01-2021	202	27-07-2022	383	15-03-2023	577
10-02-2021	330	01-04-2022	818	14-04-2023	681
30-08-2021	222	10-01-2022	407	27-03-2023	159

22-05-2021	239	17-08-2022	319	12-01-2023	356
20-07-2021	209	30-05-2022	652	14-05-2023	514
06-01-2021	130	18-08-2022	220	03-06-2023	255
15-03-2021	283	07-12-2022	291	30-09-2023	317
14-01-2021	260	20-04-2022	216	13-06-2023	606
11-07-2021	133	25-10-2022	504	07-04-2023	534
23-10-2021	159	01-05-2022	240	08-02-2023	112
24-09-2021	478	17-04-2022	711	20-05-2023	631
06-07-2021	204	22-12-2022	155	19-07-2023	297
22-04-2021	435	14-01-2022	80	25-10-2023	286
24-10-2021	501	17-06-2022	202	01-02-2023	120
17-04-2021	678	10-05-2022	402	18-09-2023	213
18-01-2021	102	22-07-2022	293	12-04-2023	996
21-12-2021	155	07-03-2022	587	05-06-2023	279
12-09-2021	174	15-04-2022	526	03-05-2023	1057
21-12-2021	97	27-03-2022	1102	06-02-2023	87
08-06-2021	495	31-07-2022	468	26-08-2023	77
04-04-2021	244	03-05-2022	275	09-02-2023	109
04-01-2021	112	01-02-2022	400	01-03-2023	127
05-03-2021	600	22-12-2022	270	11-06-2023	305
12-05-2021	379	30-05-2022	734	12-02-2023	387
02-05-2021	369	27-05-2022	247	20-05-2023	486
13-09-2021	207	09-06-2022	141	22-09-2023	193
23-06-2021	703	10-06-2022	292	14-11-2023	228
07-12-2021	396	07-05-2022	397	21-06-2023	275
01-08-2021	103	26-04-2022	379	08-08-2023	196
15-04-2021	514	01-09-2022	190	29-03-2023	669
18-06-2021	583	14-05-2022	588	18-07-2023	137
25-11-2021	112	25-09-2022	363	23-05-2023	645
12-07-2021	272	15-05-2022	244	29-08-2023	672
20-01-2021	260	13-06-2022	535	14-12-2023	100
08-05-2021	750	11-10-2022	430	24-06-2023	210
25-10-2021	195	07-09-2022	146	01-10-2023	114
16-07-2021	219	09-06-2022	459	05-08-2023	206
23-03-2021	474	12-10-2022	776	03-11-2023	207
24-10-2021	237	31-10-2022	266	25-07-2023	189
03-03-2021	528	06-06-2022	598	22-06-2023	237
18-06-2021	547	22-07-2022	137	13-09-2023	520
10-09-2021	240	13-01-2022	199	29-08-2023	306
16-06-2021	135	23-12-2022	339	05-03-2023	685
01-12-2021	283	21-07-2022	137	02-12-2023	412
27-11-2021	305	25-03-2022	811	30-10-2023	111
06-03-2021	191	01-05-2022	161	11-12-2023	232
03-04-2021	714	11-06-2022	421	16-04-2023	658
28-06-2021	444	11-03-2022	649	12-03-2023	882
13-05-2021	992	14-03-2022	203	09-08-2023	173
17-12-2021	188	23-04-2022	597	21-07-2023	285
10-12-2021	217	13-01-2022	294	08-11-2023	84
22-12-2021	125	17-08-2022	133	18-02-2023	145

13-11-2021	80	20-05-2022	628	25-10-2023	264
03-01-2021	120	31-10-2022	390	30-05-2023	522
18-10-2021	568	11-10-2022	482	11-04-2023	1039
29-03-2021	702	06-12-2022	139	27-07-2023	182
04-06-2021	366	06-09-2022	260	29-06-2023	675
14-01-2021	180	07-12-2022	138	30-09-2023	454
09-08-2021	727	22-06-2022	643	01-01-2023	220
13-06-2021	388	30-09-2022	216	31-05-2023	433
12-11-2021	171	11-10-2022	294	22-11-2023	562
26-03-2021	399	28-04-2022	315	10-01-2023	230
13-09-2021	584	19-11-2022	301	03-06-2023	166
22-03-2021	323	17-02-2022	145	30-04-2023	315
17-11-2021	458	04-01-2022	172	03-10-2023	514
20-06-2021	749	06-08-2022	137	14-04-2023	396
09-02-2021	246	19-12-2022	138	31-01-2023	355
22-01-2021	250	11-11-2022	140	08-01-2023	188
10-07-2021	273	05-05-2022	390	20-04-2023	259
22-11-2021	63	26-03-2022	803	11-06-2023	328
08-11-2021	217	13-04-2022	1026	29-01-2023	185
16-11-2021	166	29-11-2022	242	16-03-2023	429
16-05-2021	424	24-03-2022	220	29-01-2023	367
26-10-2021	356	27-07-2022	519	08-02-2023	369
02-12-2021	356	20-03-2022	641	22-07-2023	301
14-10-2021	171	01-03-2022	328	08-05-2023	127
12-09-2021	283	18-11-2022	352	10-07-2023	231
14-04-2021	165	22-02-2022	121	20-07-2023	314
24-08-2021	374	04-03-2022	406	25-07-2023	500
08-10-2021	73	27-06-2022	606	22-10-2023	364
04-12-2021	107	16-04-2022	475	12-11-2023	221
10-11-2021	224	19-05-2022	376	21-07-2023	254
13-10-2021	230	20-12-2022	399	05-11-2023	237
03-01-2021	594	10-06-2022	560	01-07-2023	316
14-12-2021	110	27-06-2022	670	13-03-2023	594
07-02-2021	383	20-07-2022	339	17-01-2023	371
12-03-2021	190	20-06-2022	685	03-08-2023	496
03-11-2021	342	23-07-2022	381	25-08-2023	432
26-09-2021	664	25-05-2022	238	09-07-2023	188

2024 – 2025 Datasets

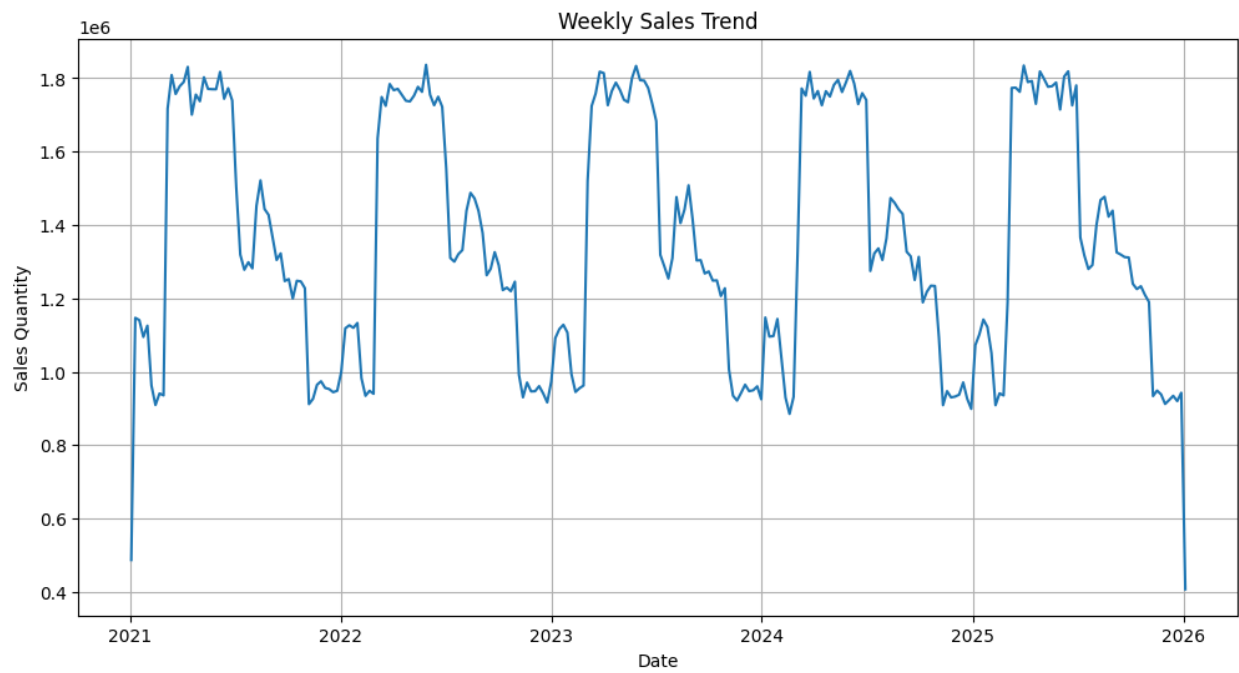
Date	Sales Qty	Date	Sales Qty
22-08-2024	381	13-05-2025	844
29-02-2024	220	07-09-2025	312
27-06-2024	292	31-10-2025	658
25-04-2024	578	14-08-2025	172
29-12-2024	212	05-09-2025	388
23-05-2024	369	25-04-2025	241
06-08-2024	281	09-10-2025	356

03-06-2024	618	15-12-2025	296
23-06-2024	873	17-10-2025	195
10-03-2024	436	30-08-2025	254
12-07-2024	684	14-05-2025	562
19-06-2024	564	08-09-2025	225
09-05-2024	537	18-03-2025	464
30-01-2024	116	12-12-2025	128
05-08-2024	113	26-09-2025	267
06-04-2024	617	26-07-2025	520
14-02-2024	384	15-11-2025	148
05-01-2024	119	28-11-2025	101
06-01-2024	532	16-05-2025	709
25-11-2024	436	27-08-2025	171
31-08-2024	237	04-02-2025	362
09-10-2024	172	12-02-2025	230
15-11-2024	342	20-04-2025	311
30-04-2024	436	17-06-2025	541
19-08-2024	536	22-10-2025	79
19-04-2024	132	08-01-2025	418
09-02-2024	191	14-04-2025	364
30-09-2024	219	21-05-2025	261
19-04-2024	207	05-10-2025	302
15-01-2024	440	25-11-2025	113
03-02-2024	273	03-05-2025	112
28-07-2024	537	05-03-2025	642
26-10-2024	339	08-05-2025	258
22-10-2024	527	20-01-2025	200
18-02-2024	383	01-11-2025	184
25-04-2024	100	24-08-2025	748
24-02-2024	116	23-04-2025	837
09-02-2024	342	27-11-2025	222
25-04-2024	670	17-03-2025	1098
11-01-2024	97	05-02-2025	172
04-05-2024	207	05-08-2025	358
26-03-2024	790	07-11-2025	222
06-09-2024	100	29-10-2025	296
18-10-2024	177	12-01-2025	135
30-07-2024	351	09-08-2025	313
04-08-2024	75	14-12-2025	152
30-09-2024	442	18-11-2025	171
20-07-2024	163	24-02-2025	128
17-04-2024	276	18-08-2025	101
22-01-2024	51	18-02-2025	117
07-09-2024	409	27-01-2025	448
10-12-2024	111	28-12-2025	529
29-03-2024	865	09-09-2025	284
19-05-2024	181	19-03-2025	316
30-01-2024	603	02-09-2025	81
12-03-2024	532	03-08-2025	178

23-07-2024	165	28-12-2025	148
13-01-2024	507	27-04-2025	355
14-02-2024	373	05-03-2025	522
02-02-2024	350	25-09-2025	106
03-07-2024	338	18-01-2025	212
22-05-2024	442	12-04-2025	505
10-07-2024	270	20-09-2025	285
10-01-2024	239	02-12-2025	360
10-11-2024	386	07-12-2025	138
14-06-2024	286	16-05-2025	255
04-09-2024	530	08-03-2025	367
28-10-2024	308	21-11-2025	229
15-09-2024	224	28-01-2025	223
24-11-2024	453	21-03-2025	336
19-10-2024	505	11-04-2025	442
07-01-2024	538	06-02-2025	370
25-06-2024	430	10-10-2025	416
29-03-2024	630	10-11-2025	241
30-04-2024	685	08-12-2025	530
18-03-2024	198	11-12-2025	425
23-09-2024	168	06-08-2025	168
03-03-2024	438	05-09-2025	389
09-10-2024	486	05-03-2025	334
22-09-2024	372	03-03-2025	822
13-12-2024	150	20-05-2025	567
20-12-2024	580	27-10-2025	132
04-02-2024	266	20-05-2025	318
03-01-2024	129	12-06-2025	882
11-02-2024	165	01-11-2025	92
05-02-2024	126	27-06-2025	372
26-06-2024	843	08-10-2025	273
18-02-2024	188	07-12-2025	82
22-02-2024	112	18-05-2025	693
25-10-2024	225	19-07-2025	508
10-01-2024	163	26-08-2025	765
25-04-2024	469	10-02-2025	323
04-11-2024	392	07-02-2025	176
18-05-2024	609	29-03-2025	670
22-03-2024	761	21-08-2025	271
29-06-2024	748	19-01-2025	279
16-09-2024	497	23-07-2025	219
10-07-2024	376	01-06-2025	670
31-05-2024	417	21-05-2025	307

Source: Podaran Foods India Private Limited

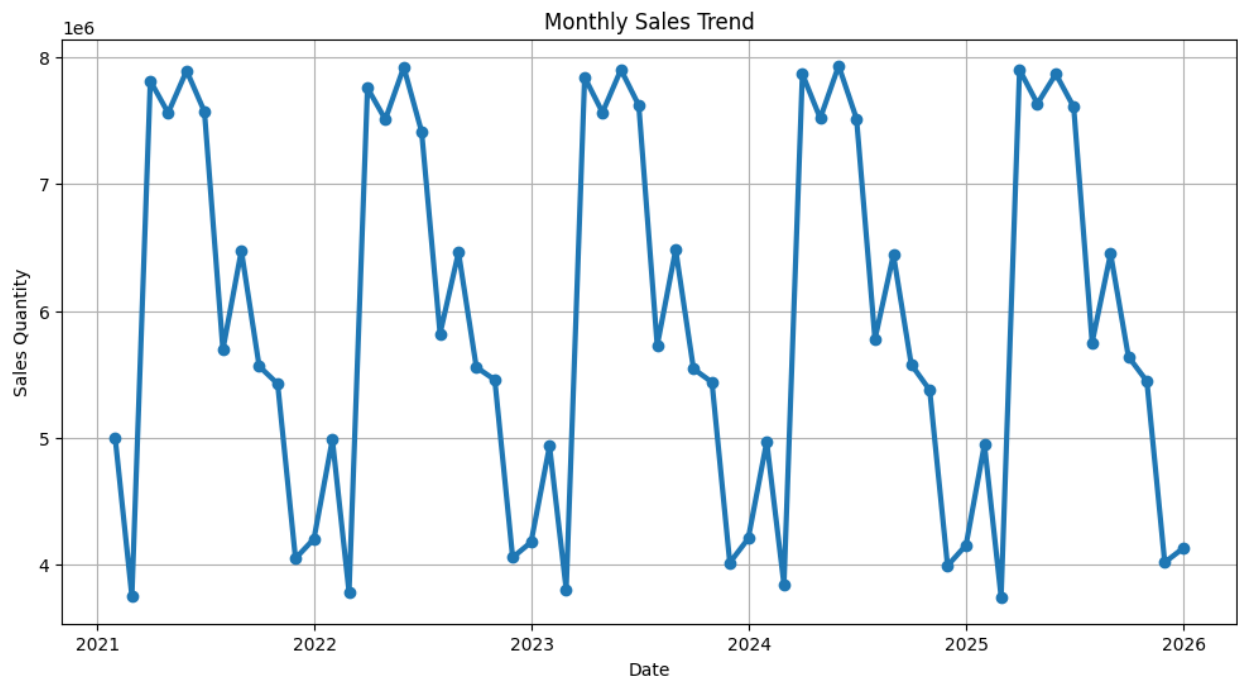
4.5.1 Weekly Sales Trend



Interpretation:

The above graph shows the weekly sales trend. It provides a smoother representation of data compared to daily values and helps in identifying short-term fluctuations in demand. The variations indicate changes in customer purchasing behavior across different weeks.

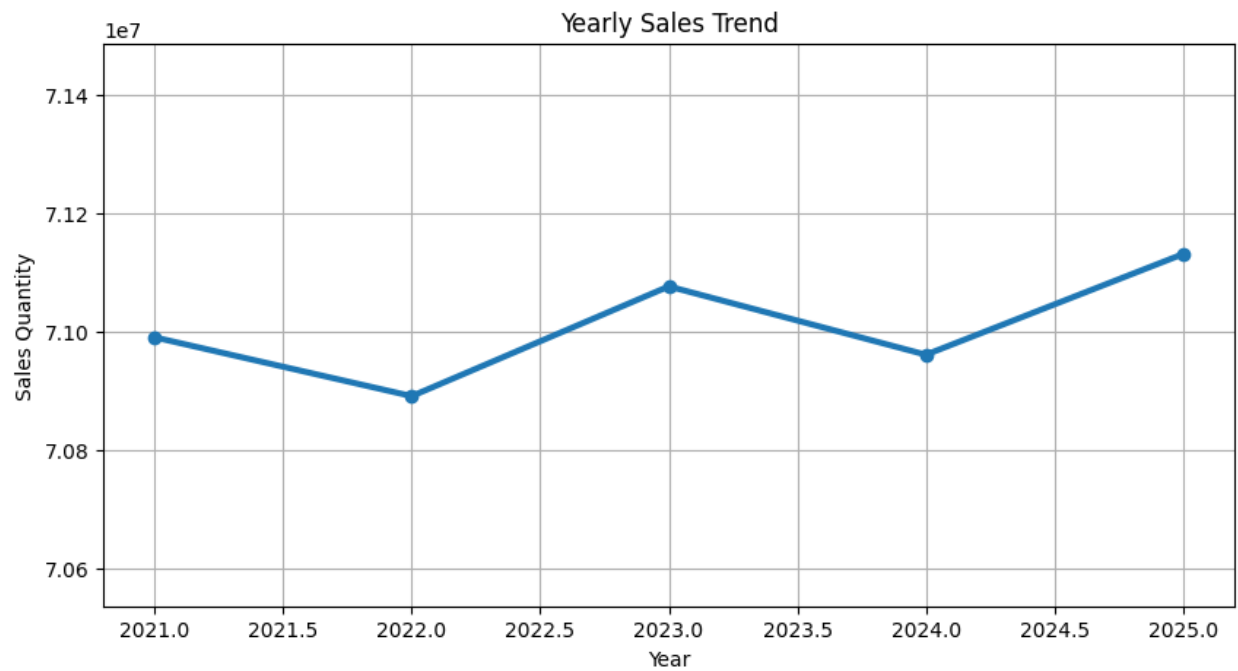
4.5.2 Monthly Sales Trend



Interpretation:

The graph shows the monthly sales trend of juice products over the study period. It is observed that sales fluctuate across different months, indicating seasonal variations. Higher sales are noticed during summer months, while lower sales occur during off-season periods.

4.5.3 Yearly Sales Trend



Interpretation:

The yearly trend indicates overall business performance across years. It shows gradual growth with slight fluctuations, suggesting stable demand and consistent market performance.

4.6 Business Analysis

Table no 4.5.2 (2021 – 2023 Datasets)

Region	Sales_Qty	Region	Sales_Qty	Region	Sales_Qty
Tamil Nadu	333	Kerala	370	Tamil Nadu	450
Tamil Nadu	495	Tamil Nadu	547	Tamil Nadu	220
Tamil Nadu	267	Kerala	278	Kerala	110
Tamil Nadu	246	Tamil Nadu	777	Tamil Nadu	393
Tamil Nadu	262	Karnataka	669	Andhra Pradesh	108
Kerala	199	Tamil Nadu	579	Karnataka	481
Kerala	675	Tamil Nadu	818	Tamil Nadu	801
Kerala	115	Tamil Nadu	564	Karnataka	240
Tamil Nadu	282	Tamil Nadu	342	Andhra Pradesh	168
Kerala	516	Kerala	312	Karnataka	233
Kerala	202	Tamil Nadu	383	Karnataka	577
Tamil Nadu	330	Kerala	818	Kerala	681
Karnataka	222	Tamil Nadu	407	Tamil Nadu	159
Andhra Pradesh	239	Tamil Nadu	319	Kerala	356
Kerala	209	Tamil Nadu	652	Kerala	514
Andhra Pradesh	130	Tamil Nadu	220	Tamil Nadu	255
Andhra Pradesh	283	Tamil Nadu	291	Tamil Nadu	317
Kerala	260	Tamil Nadu	216	Kerala	606
Tamil Nadu	133	Karnataka	504	Tamil Nadu	534
Tamil Nadu	159	Kerala	240	Kerala	112
Kerala	478	Karnataka	711	Karnataka	631
Andhra Pradesh	204	Tamil Nadu	155	Andhra Pradesh	297
Tamil Nadu	435	Kerala	80	Karnataka	286
Tamil Nadu	501	Kerala	202	Kerala	120
Karnataka	678	Tamil Nadu	402	Karnataka	213
Kerala	102	Andhra Pradesh	293	Tamil Nadu	996
Andhra Pradesh	155	Tamil Nadu	587	Tamil Nadu	279
Tamil Nadu	174	Kerala	526	Tamil Nadu	1057
Tamil Nadu	97	Tamil Nadu	1102	Karnataka	87
Tamil Nadu	495	Tamil Nadu	468	Tamil Nadu	77
Andhra Pradesh	244	Andhra Pradesh	275	Tamil Nadu	109
Kerala	112	Karnataka	400	Andhra Pradesh	127
Kerala	600	Tamil Nadu	270	Kerala	305
Tamil Nadu	379	Tamil Nadu	734	Tamil Nadu	387

Andhra Pradesh	369	Kerala	247	Kerala	486
Karnataka	207	Andhra Pradesh	141	Kerala	193
Kerala	703	Tamil Nadu	292	Kerala	228
Kerala	396	Karnataka	397	Kerala	275
Tamil Nadu	103	Kerala	379	Tamil Nadu	196
Karnataka	514	Tamil Nadu	190	Tamil Nadu	669
Kerala	583	Karnataka	588	Tamil Nadu	137
Andhra Pradesh	112	Karnataka	363	Tamil Nadu	645
Andhra Pradesh	272	Tamil Nadu	244	Kerala	672
Tamil Nadu	260	Tamil Nadu	535	Kerala	100
Tamil Nadu	750	Tamil Nadu	430	Tamil Nadu	210
Karnataka	195	Andhra Pradesh	146	Kerala	114
Kerala	219	Kerala	459	Tamil Nadu	206
Tamil Nadu	474	Tamil Nadu	776	Kerala	207
Tamil Nadu	237	Kerala	266	Tamil Nadu	189
Tamil Nadu	528	Tamil Nadu	598	Kerala	237
Andhra Pradesh	547	Andhra Pradesh	137	Karnataka	520
Kerala	240	Tamil Nadu	199	Tamil Nadu	306
Andhra Pradesh	135	Karnataka	339	Karnataka	685
Kerala	283	Tamil Nadu	137	Tamil Nadu	412
Tamil Nadu	305	Tamil Nadu	811	Andhra Pradesh	111
Tamil Nadu	191	Andhra Pradesh	161	Tamil Nadu	232
Karnataka	714	Tamil Nadu	421	Tamil Nadu	658
Tamil Nadu	444	Karnataka	649	Tamil Nadu	882
Tamil Nadu	992	Tamil Nadu	203	Tamil Nadu	173
Tamil Nadu	188	Tamil Nadu	597	Tamil Nadu	285
Kerala	217	Tamil Nadu	294	Karnataka	84
Kerala	125	Karnataka	133	Kerala	145
Kerala	80	Andhra Pradesh	628	Tamil Nadu	264
Karnataka	120	Tamil Nadu	390	Tamil Nadu	522
Kerala	568	Kerala	482	Tamil Nadu	1039
Tamil Nadu	702	Tamil Nadu	139	Tamil Nadu	182
Andhra Pradesh	366	Andhra Pradesh	260	Kerala	675
Tamil Nadu	180	Kerala	138	Karnataka	454
Tamil Nadu	727	Kerala	643	Karnataka	220
Andhra Pradesh	388	Andhra Pradesh	216	Karnataka	433

Andhra Pradesh	171	Andhra Pradesh	294	Tamil Nadu	562
Karnataka	399	Tamil Nadu	315	Kerala	230
Tamil Nadu	584	Kerala	301	Kerala	166
Tamil Nadu	323	Andhra Pradesh	145	Tamil Nadu	315
Andhra Pradesh	458	Kerala	172	Tamil Nadu	514
Tamil Nadu	749	Andhra Pradesh	137	Tamil Nadu	396
Tamil Nadu	246	Tamil Nadu	138	Karnataka	355
Kerala	250	Karnataka	140	Karnataka	188
Andhra Pradesh	273	Tamil Nadu	390	Tamil Nadu	259
Tamil Nadu	63	Tamil Nadu	803	Tamil Nadu	328
Tamil Nadu	217	Tamil Nadu	1026	Tamil Nadu	185
Kerala	166	Karnataka	242	Kerala	429
Tamil Nadu	424	Karnataka	220	Tamil Nadu	367
Tamil Nadu	356	Karnataka	519	Tamil Nadu	369
Tamil Nadu	356	Kerala	641	Karnataka	301
Kerala	171	Tamil Nadu	328	Tamil Nadu	127
Tamil Nadu	283	Kerala	352	Andhra Pradesh	231
Karnataka	165	Tamil Nadu	121	Karnataka	314
Kerala	374	Tamil Nadu	406	Kerala	500
Andhra Pradesh	73	Tamil Nadu	606	Karnataka	364
Karnataka	107	Tamil Nadu	475	Tamil Nadu	221
Karnataka	224	Tamil Nadu	376	Karnataka	254
Tamil Nadu	230	Kerala	399	Andhra Pradesh	237
Tamil Nadu	594	Tamil Nadu	560	Tamil Nadu	316
Kerala	110	Kerala	670	Karnataka	594
Karnataka	383	Kerala	339	Karnataka	371
Kerala	190	Tamil Nadu	685	Tamil Nadu	496
Kerala	342	Tamil Nadu	381	Tamil Nadu	432
Kerala	664	Kerala	238	Tamil Nadu	188

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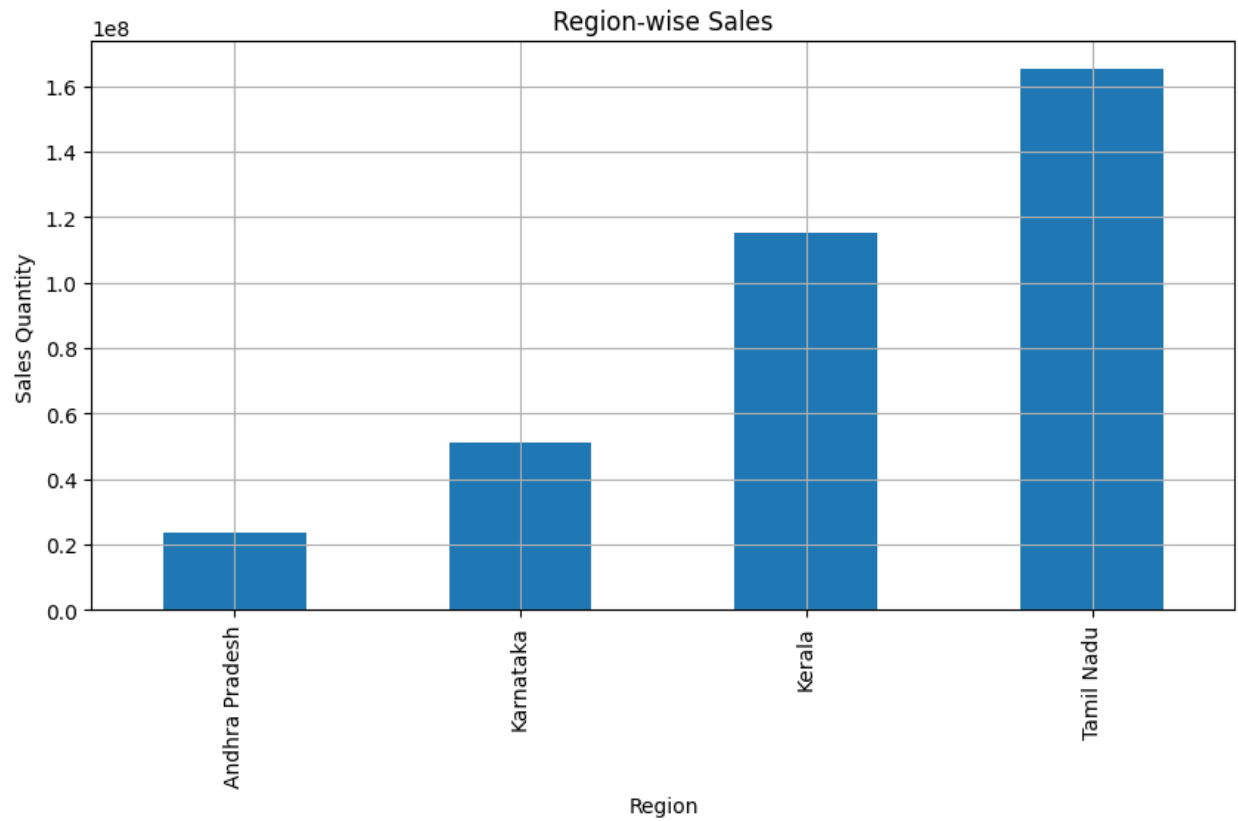
Region	Sales Qty	Region	Sales Qty
Tamil Nadu	381	Tamil Nadu	844
Kerala	220	Karnataka	312
Kerala	292	Tamil Nadu	658
Tamil Nadu	578	Tamil Nadu	172
Tamil Nadu	212	Karnataka	388
Kerala	369	Tamil Nadu	241
Tamil Nadu	281	Karnataka	356

Tamil Nadu	618	Karnataka	296
Tamil Nadu	873	Karnataka	195
Tamil Nadu	436	Karnataka	254
Tamil Nadu	684	Kerala	562
Kerala	564	Kerala	225
Kerala	537	Tamil Nadu	464
Karnataka	116	Tamil Nadu	128
Andhra Pradesh	113	Kerala	267
Tamil Nadu	617	Tamil Nadu	520
Tamil Nadu	384	Karnataka	148
Tamil Nadu	119	Kerala	101
Kerala	532	Tamil Nadu	709
Kerala	436	Tamil Nadu	171
Tamil Nadu	237	Kerala	362
Tamil Nadu	172	Kerala	230
Tamil Nadu	342	Kerala	311
Karnataka	436	Karnataka	541
Tamil Nadu	536	Tamil Nadu	79
Andhra Pradesh	132	Kerala	418
Tamil Nadu	191	Tamil Nadu	364
Andhra Pradesh	219	Karnataka	261
Tamil Nadu	207	Tamil Nadu	302
Tamil Nadu	440	Andhra Pradesh	113
Kerala	273	Andhra Pradesh	112
Karnataka	537	Tamil Nadu	642
Tamil Nadu	339	Kerala	258
Tamil Nadu	527	Kerala	200
Karnataka	383	Andhra Pradesh	184
Andhra Pradesh	100	Kerala	748
Karnataka	116	Tamil Nadu	837
Kerala	342	Tamil Nadu	222
Tamil Nadu	670	Kerala	1098
Kerala	97	Kerala	172
Tamil Nadu	207	Kerala	358
Tamil Nadu	790	Tamil Nadu	222
Tamil Nadu	100	Andhra Pradesh	296
Tamil Nadu	177	Tamil Nadu	135
Tamil Nadu	351	Tamil Nadu	313
Andhra Pradesh	75	Tamil Nadu	152
Kerala	442	Tamil Nadu	171
Tamil Nadu	163	Tamil Nadu	128
Tamil Nadu	276	Andhra Pradesh	101
Andhra Pradesh	51	Kerala	117
Tamil Nadu	409	Kerala	448
Kerala	111	Tamil Nadu	529
Tamil Nadu	865	Tamil Nadu	284
Tamil Nadu	181	Andhra Pradesh	316
Tamil Nadu	603	Tamil Nadu	81
Karnataka	532	Tamil Nadu	178

Kerala	165	Kerala	148
Tamil Nadu	507	Tamil Nadu	355
Kerala	373	Tamil Nadu	522
Kerala	350	Andhra Pradesh	106
Tamil Nadu	338	Kerala	212
Kerala	442	Tamil Nadu	505
Tamil Nadu	270	Tamil Nadu	285
Andhra Pradesh	239	Kerala	360
Kerala	386	Tamil Nadu	138
Tamil Nadu	286	Karnataka	255
Andhra Pradesh	530	Tamil Nadu	367
Andhra Pradesh	308	Karnataka	229
Tamil Nadu	224	Kerala	223
Kerala	453	Kerala	336
Kerala	505	Kerala	442
Kerala	538	Kerala	370
Tamil Nadu	430	Kerala	416
Andhra Pradesh	630	Tamil Nadu	241
Tamil Nadu	685	Tamil Nadu	530
Kerala	198	Tamil Nadu	425
Tamil Nadu	168	Andhra Pradesh	168
Kerala	438	Karnataka	389
Karnataka	486	Kerala	334
Kerala	372	Tamil Nadu	822
Karnataka	150	Karnataka	567
Kerala	580	Kerala	132
Tamil Nadu	266	Karnataka	318
Andhra Pradesh	129	Tamil Nadu	882
Karnataka	165	Tamil Nadu	92
Tamil Nadu	126	Kerala	372
Tamil Nadu	843	Kerala	273
Kerala	188	Tamil Nadu	82
Kerala	112	Tamil Nadu	693
Tamil Nadu	225	Kerala	508
Kerala	163	Kerala	765
Tamil Nadu	469	Tamil Nadu	323
Kerala	392	Kerala	176
Kerala	609	Karnataka	670
Tamil Nadu	761	Tamil Nadu	271
Tamil Nadu	748	Tamil Nadu	279
Tamil Nadu	497	Andhra Pradesh	219
Kerala	376	Kerala	670
Tamil Nadu	417	Kerala	307

Source: Podaran Foods India Private Limited

4.6.1 Region-wise Sales



Interpretation:

The graph shows the distribution of sales across different regions. It helps in identifying high-demand regions and understanding geographical variations in sales performance. This chart represents the highest region wise sales is Tamil Nadu and Kerala.

4.6.2 Product-wise Sales

Table no 4.5.3 (2021 – 2023 Datasets)

Product_Name	Sales_Qty	Product_Name	Sales_Qty	Product_Name	Sales_Qty
Mixed Fruit	333	Cola	370	Jeera Masala	450
Mango	495	Cola	547	Mixed Fruit	220
Pista	267	Mango	278	Mango	110
Orange	246	Paneer Soda	777	Chocolate	393
Apple	262	Cola	669	Orange	108
Pista	199	Mixed Fruit	579	Mango	481
Apple	675	Mango	818	Mixed Fruit	801
Pista	115	Mango	564	Cola	240
Orange	282	Pista	342	Badam	168
Mango	516	Orange	312	Cola	233
Strawberry	202	Orange	383	Chocolate	577
Jeera Masala	330	Mixed Fruit	818	Mixed Fruit	681
Chocolate	222	Clear Lemon	407	Badam	159
Mixed Fruit	239	Apple	319	Badam	356
Mixed Fruit	209	Jeera Masala	652	Apple	514
Pista	130	Jeera Masala	220	Pista	255
Paneer Soda	283	Orange	291	Jeera Masala	317
Orange	260	Chocolate	216	Apple	606
Strawberry	133	Badam	504	Mixed Fruit	534
Pista	159	Strawberry	240	Orange	112
Cola	478	Pista	711	Clear Lemon	631
Mixed Fruit	204	Pista	155	Jeera Masala	297
Jeera Masala	435	Strawberry	80	Pista	286
Jeera Masala	501	Mixed Fruit	202	Paneer Soda	120
Mixed Fruit	678	Cola	402	Cola	213
Cola	102	Orange	293	Mango	996
Badam	155	Paneer Soda	587	Paneer Soda	279
Paneer Soda	174	Chocolate	526	Mango	1057
Strawberry	97	Cola	1102	Mixed Fruit	87
Strawberry	495	Cola	468	Badam	77
Chocolate	244	Jeera Masala	275	Clear Lemon	109
Paneer Soda	112	Strawberry	400	Jeera Masala	127
Paneer Soda	600	Paneer Soda	270	Mango	305
Pista	379	Mixed Fruit	734	Clear Lemon	387
Pista	369	Chocolate	247	Cola	486
Strawberry	207	Chocolate	141	Mango	193
Chocolate	703	Jeera Masala	292	Paneer Soda	228
Pista	396	Clear Lemon	397	Mango	275
Badam	103	Mango	379	Strawberry	196
Mixed Fruit	514	Jeera Masala	190	Orange	669
Orange	583	Pista	588	Mixed Fruit	137
Pista	112	Mango	363	Jeera Masala	645
Jeera Masala	272	Badam	244	Jeera Masala	672
Mango	260	Paneer Soda	535	Paneer Soda	100

Mixed Fruit	750	Orange	430	Chocolate	210
Pista	195	Cola	146	Badam	114
Jeera Masala	219	Pista	459	Clear Lemon	206
Strawberry	474	Paneer Soda	776	Pista	207
Jeera Masala	237	Apple	266	Badam	189
Paneer Soda	528	Paneer Soda	598	Pista	237
Mixed Fruit	547	Clear Lemon	137	Paneer Soda	520
Apple	240	Strawberry	199	Cola	306
Strawberry	135	Apple	339	Strawberry	685
Chocolate	283	Strawberry	137	Apple	412
Mixed Fruit	305	Mixed Fruit	811	Chocolate	111
Badam	191	Orange	161	Strawberry	232
Strawberry	714	Badam	421	Clear Lemon	658
Mixed Fruit	444	Orange	649	Apple	882
Paneer Soda	992	Strawberry	203	Orange	173
Strawberry	188	Clear Lemon	597	Pista	285
Strawberry	217	Orange	294	Apple	84
Badam	125	Chocolate	133	Chocolate	145
Chocolate	80	Mango	628	Apple	264
Badam	120	Mixed Fruit	390	Clear Lemon	522
Mango	568	Cola	482	Mango	1039
Mixed Fruit	702	Chocolate	139	Orange	182
Paneer Soda	366	Pista	260	Jeera Masala	675
Strawberry	180	Pista	138	Mango	454
Mango	727	Jeera Masala	643	Clear Lemon	220
Cola	388	Mixed Fruit	216	Pista	433
Pista	171	Mango	294	Mango	562
Strawberry	399	Clear Lemon	315	Cola	230
Mixed Fruit	584	Paneer Soda	301	Badam	166
Chocolate	323	Pista	145	Mango	315
Mango	458	Mixed Fruit	172	Apple	514
Paneer Soda	749	Mango	137	Orange	396
Badam	246	Badam	138	Cola	355
Orange	250	Orange	140	Jeera Masala	188
Jeera Masala	273	Apple	390	Apple	259
Badam	63	Mango	803	Strawberry	328
Mixed Fruit	217	Mango	1026	Chocolate	185
Paneer Soda	166	Orange	242	Mixed Fruit	429
Clear Lemon	424	Badam	220	Badam	367
Chocolate	356	Mango	519	Jeera Masala	369
Clear Lemon	356	Clear Lemon	641	Cola	301
Jeera Masala	171	Badam	328	Badam	127
Strawberry	283	Clear Lemon	352	Strawberry	231
Cola	165	Strawberry	121	Strawberry	314
Apple	374	Clear Lemon	406	Paneer Soda	500
Orange	73	Jeera Masala	606	Strawberry	364
Mixed Fruit	107	Badam	475	Apple	221
Badam	224	Orange	376	Orange	254
Badam	230	Apple	399	Cola	237

Mango	594	Mango	560	Cola	316
Orange	110	Strawberry	670	Clear Lemon	594
Strawberry	383	Apple	339	Mango	371
Chocolate	190	Jeera Masala	685	Jeera Masala	496
Chocolate	342	Paneer Soda	381	Orange	432
Jeera Masala	664	Chocolate	238	Chocolate	188

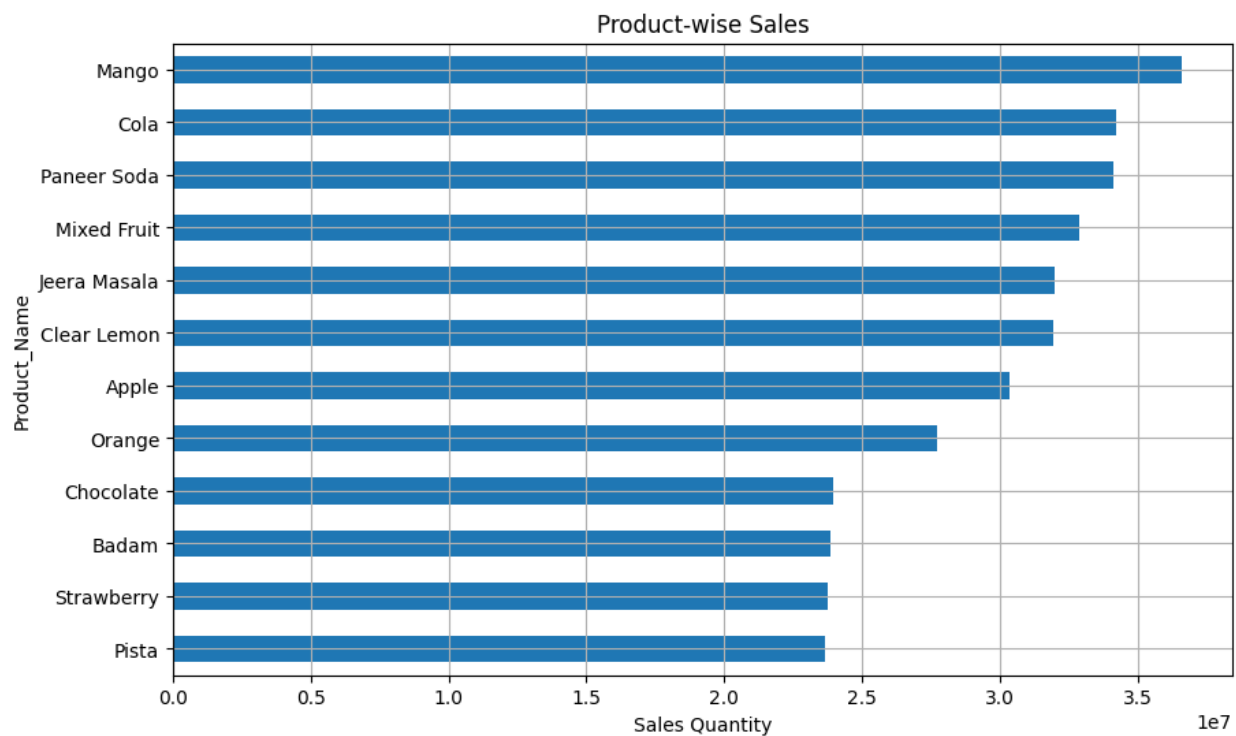
2024 – 2025 Datasets

Product Name	Sales Qty	Product Name	Sales Qty
Cola	381	Mixed Fruit	844
Mixed Fruit	220	Badam	312
Chocolate	292	Mango	658
Mango	578	Clear Lemon	172
Badam	212	Strawberry	388
Badam	369	Pista	241
Apple	281	Orange	356
Paneer Soda	618	Paneer Soda	296
Apple	873	Clear Lemon	195
Clear Lemon	436	Paneer Soda	254
Paneer Soda	684	Badam	562
Mango	564	Orange	225
Orange	537	Strawberry	464
Chocolate	116	Mixed Fruit	128
Paneer Soda	113	Orange	267
Mixed Fruit	617	Mixed Fruit	520
Orange	384	Chocolate	148
Strawberry	119	Cola	101
Clear Lemon	532	Jeera Masala	709
Clear Lemon	436	Clear Lemon	171
Mixed Fruit	237	Cola	362
Orange	172	Mango	230
Jeera Masala	342	Mango	311
Chocolate	436	Jeera Masala	541
Orange	536	Strawberry	79
Cola	132	Mixed Fruit	418
Strawberry	191	Orange	364
Clear Lemon	219	Orange	261
Pista	207	Mixed Fruit	302
Mango	440	Pista	113
Chocolate	273	Apple	112
Pista	537	Jeera Masala	642
Pista	339	Clear Lemon	258
Cola	527	Strawberry	200
Mixed Fruit	383	Jeera Masala	184
Strawberry	100	Mango	748
Strawberry	116	Cola	837
Paneer Soda	342	Jeera Masala	222

Jeera Masala	670	Jeera Masala	1098
Chocolate	97	Orange	172
Clear Lemon	207	Apple	358
Apple	790	Jeera Masala	222
Strawberry	100	Clear Lemon	296
Mango	177	Mixed Fruit	135
Paneer Soda	351	Chocolate	313
Pista	75	Orange	152
Chocolate	442	Pista	171
Clear Lemon	163	Mixed Fruit	128
Jeera Masala	276	Apple	101
Cola	51	Badam	117
Mixed Fruit	409	Jeera Masala	448
Chocolate	111	Paneer Soda	529
Apple	865	Pista	284
Chocolate	181	Jeera Masala	316
Cola	603	Pista	81
Jeera Masala	532	Jeera Masala	178
Clear Lemon	165	Apple	148
Clear Lemon	507	Orange	355
Orange	373	Chocolate	522
Mixed Fruit	350	Chocolate	106
Jeera Masala	338	Mixed Fruit	212
Mango	442	Pista	505
Mixed Fruit	270	Apple	285
Jeera Masala	239	Cola	360
Jeera Masala	386	Pista	138
Chocolate	286	Clear Lemon	255
Mixed Fruit	530	Clear Lemon	367
Apple	308	Strawberry	229
Chocolate	224	Strawberry	223
Jeera Masala	453	Mixed Fruit	336
Badam	505	Pista	442
Clear Lemon	538	Mango	370
Clear Lemon	430	Pista	416
Mixed Fruit	630	Chocolate	241
Orange	685	Mango	530
Pista	198	Mixed Fruit	425
Pista	168	Paneer Soda	168
Clear Lemon	438	Chocolate	389
Strawberry	486	Apple	334
Strawberry	372	Mixed Fruit	822
Chocolate	150	Apple	567
Jeera Masala	580	Mango	132
Orange	266	Strawberry	318
Pista	129	Paneer Soda	882
Pista	165	Strawberry	92
Chocolate	126	Pista	372
Paneer Soda	843	Apple	273

Mixed Fruit	188	Jeera Masala	82
Paneer Soda	112	Cola	693
Jeera Masala	225	Orange	508
Chocolate	163	Mango	765
Chocolate	469	Mixed Fruit	323
Badam	392	Pista	176
Chocolate	609	Strawberry	670
Apple	761	Chocolate	271
Mixed Fruit	748	Badam	279
Jeera Masala	497	Mango	219
Cola	376	Clear Lemon	670
Apple	417	Paneer Soda	307

Source: Podaran Foods India Private Limited



Interpretation:

This graph highlights the sales performance of various products. It helps in identifying best-selling products and those with lower demand, which supports decision-making in production planning. The Mango and cola are the highest selling product over the other juices1.

4.6.3 Bottle Size Distribution (Pie Chart)

Table no 4.5.4 (2021 – 2023 Datasets)

Bottle_Size	Sales_Qty	Bottle_Size	Sales_Qty	Bottle_Size	Sales_Qty
1.5L	333	200ml	370	200ml	450
200ml	495	200ml	547	1L	220
200ml	267	200ml	278	200ml	110
200ml	246	1.5L	777	200ml	393
200ml	262	1L	669	200ml	108
200ml	199	200ml	579	200ml	481
200ml	675	1L	818	200ml	801
1L	115	200ml	564	200ml	240
200ml	282	200ml	342	200ml	168
200ml	516	1L	312	1L	233
200ml	202	200ml	383	200ml	577
200ml	330	200ml	818	200ml	681
200ml	222	200ml	407	200ml	159
200ml	239	200ml	319	200ml	356
200ml	209	200ml	652	1L	514
200ml	130	200ml	220	200ml	255
1L	283	200ml	291	1L	317
200ml	260	200ml	216	200ml	606
200ml	133	200ml	504	200ml	534
200ml	159	200ml	240	200ml	112
1.5L	478	200ml	711	200ml	631
200ml	204	200ml	155	200ml	297
200ml	435	200ml	80	200ml	286
200ml	501	200ml	202	200ml	120
1.5L	678	200ml	402	200ml	213
200ml	102	200ml	293	200ml	996
200ml	155	200ml	587	200ml	279
1L	174	200ml	526	200ml	1057
200ml	97	200ml	1102	200ml	87
1L	495	200ml	468	200ml	77
200ml	244	200ml	275	1.5L	109
200ml	112	200ml	400	1.5L	127
200ml	600	200ml	270	200ml	305
200ml	379	200ml	734	200ml	387
200ml	369	200ml	247	200ml	486
1L	207	200ml	141	200ml	193
200ml	703	200ml	292	1L	228
1L	396	200ml	397	200ml	275
200ml	103	1L	379	200ml	196
200ml	514	200ml	190	200ml	669
1L	583	1.5L	588	200ml	137
200ml	112	200ml	363	200ml	645
200ml	272	200ml	244	200ml	672
200ml	260	200ml	535	200ml	100

1L	750	200ml	430	1L	210
200ml	195	1.5L	146	1L	114
200ml	219	200ml	459	200ml	206
200ml	474	200ml	776	200ml	207
200ml	237	200ml	266	200ml	189
200ml	528	200ml	598	200ml	237
200ml	547	200ml	137	200ml	520
1L	240	1L	199	200ml	306
200ml	135	200ml	339	1L	685
200ml	283	200ml	137	200ml	412
200ml	305	1L	811	200ml	111
200ml	191	200ml	161	200ml	232
1L	714	1L	421	200ml	658
200ml	444	200ml	649	200ml	882
1L	992	200ml	203	200ml	173
1L	188	1L	597	200ml	285
200ml	217	200ml	294	200ml	84
1.5L	125	1L	133	200ml	145
200ml	80	200ml	628	200ml	264
200ml	120	1L	390	200ml	522
200ml	568	200ml	482	200ml	1039
1L	702	1L	139	200ml	182
200ml	366	200ml	260	200ml	675
200ml	180	200ml	138	1L	454
200ml	727	1L	643	1L	220
200ml	388	200ml	216	200ml	433
200ml	171	200ml	294	1L	562
200ml	399	1L	315	200ml	230
200ml	584	200ml	301	200ml	166
200ml	323	200ml	145	200ml	315
200ml	458	200ml	172	200ml	514
200ml	749	200ml	137	200ml	396
200ml	246	200ml	138	200ml	355
200ml	250	200ml	140	200ml	188
200ml	273	1L	390	1L	259
200ml	63	200ml	803	200ml	328
200ml	217	200ml	1026	200ml	185
200ml	166	200ml	242	200ml	429
200ml	424	200ml	220	200ml	367
1.5L	356	200ml	519	200ml	369
200ml	356	1L	641	200ml	301
1.5L	171	200ml	328	200ml	127
200ml	283	200ml	352	200ml	231
200ml	165	1L	121	1.5L	314
1L	374	200ml	406	200ml	500
200ml	73	1L	606	1L	364
200ml	107	200ml	475	200ml	221
200ml	224	1L	376	1L	254
200ml	230	200ml	399	200ml	237

200ml	594	1.5L	560	1L	316
200ml	110	200ml	670	200ml	594
200ml	383	200ml	339	200ml	371
200ml	190	200ml	685	200ml	496
1L	342	200ml	381	200ml	432
200ml	664	200ml	238	1L	188

2024 – 2025 Datasets

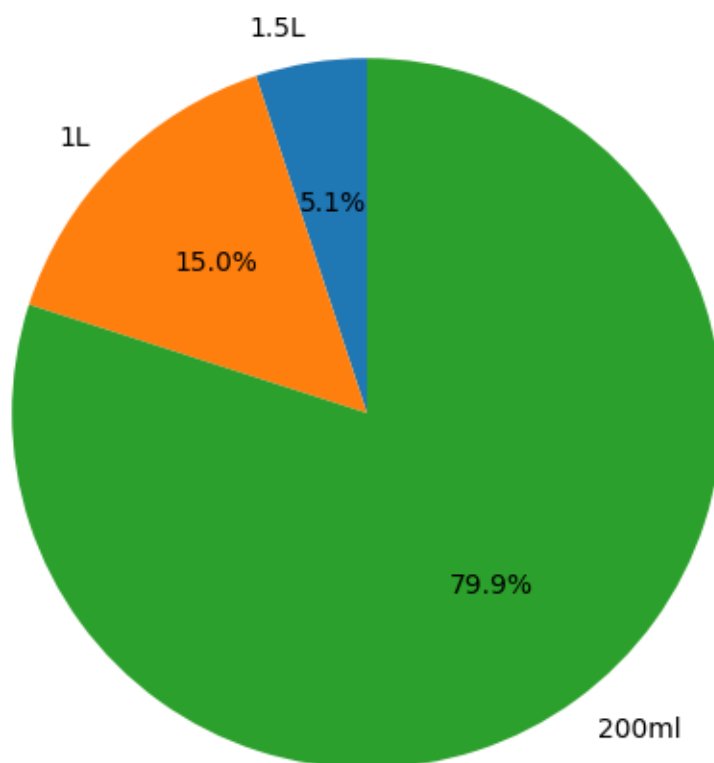
Bottle Size	Sales Qty	Bottle Size	Sales Qty
200ml	381	200ml	844
200ml	220	200ml	312
200ml	292	200ml	658
200ml	578	1L	172
200ml	212	200ml	388
200ml	369	200ml	241
200ml	281	1.5L	356
1L	618	200ml	296
1.5L	873	200ml	195
200ml	436	200ml	254
200ml	684	1L	562
200ml	564	200ml	225
200ml	537	200ml	464
1L	116	200ml	128
200ml	113	200ml	267
200ml	617	200ml	520
200ml	384	1L	148
200ml	119	200ml	101
200ml	532	200ml	709
200ml	436	200ml	171
200ml	237	200ml	362
200ml	172	1.5L	230
200ml	342	200ml	311
1L	436	200ml	541
200ml	536	200ml	79
200ml	132	1L	418
1L	191	200ml	364
200ml	219	200ml	261
1L	207	200ml	302
200ml	440	200ml	113
200ml	273	1L	112
200ml	537	200ml	642
200ml	339	1L	258
200ml	527	200ml	200
200ml	383	200ml	184
200ml	100	1L	748
200ml	116	1L	837
200ml	342	200ml	222

200ml	670	200ml	1098
200ml	97	200ml	172
200ml	207	200ml	358
200ml	790	200ml	222
200ml	100	1L	296
200ml	177	200ml	135
200ml	351	200ml	313
200ml	75	1L	152
200ml	442	200ml	171
1L	163	200ml	128
200ml	276	200ml	101
200ml	51	1L	117
200ml	409	200ml	448
200ml	111	200ml	529
200ml	865	200ml	284
200ml	181	200ml	316
200ml	603	200ml	81
1L	532	200ml	178
200ml	165	200ml	148
200ml	507	200ml	355
200ml	373	1L	522
200ml	350	200ml	106
200ml	338	1L	212
1.5L	442	1L	505
200ml	270	200ml	285
200ml	239	200ml	360
200ml	386	200ml	138
200ml	286	1L	255
200ml	530	200ml	367
200ml	308	200ml	229
200ml	224	200ml	223
200ml	453	200ml	336
200ml	505	200ml	442
200ml	538	1L	370
200ml	430	200ml	416
200ml	630	200ml	241
200ml	685	200ml	530
200ml	198	200ml	425
200ml	168	200ml	168
200ml	438	200ml	389
200ml	486	200ml	334
1L	372	200ml	822
200ml	150	1L	567
200ml	580	200ml	132
200ml	266	200ml	318
200ml	129	200ml	882
200ml	165	1L	92
200ml	126	200ml	372
1L	843	200ml	273

1L	188	200ml	82
200ml	112	200ml	693
200ml	225	200ml	508
200ml	163	200ml	765
200ml	469	200ml	323
200ml	392	200ml	176
200ml	609	200ml	670
200ml	761	200ml	271
1.5L	748	200ml	279
200ml	497	200ml	219
200ml	376	200ml	670
200ml	417	200ml	307

Source: Podaran Foods India Private Limited

Sales Distribution by Bottle Size



Interpretation:

The pie chart represents the percentage contribution of different bottle sizes to total sales. It clearly shows customer preference towards specific packaging sizes. It indicates the 200ml is contribute more in the sales over other bottles.

4.6.4 Seasonal Analysis

Table no 4.5.5 (2021 -2023 Datasets)

Date	Sales_Qty	Date	Sales_Qty	Date	Sales_Qty
02-11-2021	333	14-11-2022	370	19-10-2023	450
05-11-2021	495	16-02-2022	547	30-10-2023	220
10-03-2021	267	27-01-2022	278	06-02-2023	110
03-09-2021	246	02-01-2022	777	23-04-2023	393
19-01-2021	262	31-05-2022	669	20-06-2023	108
28-06-2021	199	30-09-2022	579	12-06-2023	481
05-05-2021	675	21-08-2022	818	03-03-2023	801
23-12-2021	115	12-08-2022	564	30-05-2023	240
21-12-2021	282	24-01-2022	342	30-12-2023	168
28-05-2021	516	17-07-2022	312	27-07-2023	233
14-01-2021	202	27-07-2022	383	15-03-2023	577
10-02-2021	330	01-04-2022	818	14-04-2023	681
30-08-2021	222	10-01-2022	407	27-03-2023	159
22-05-2021	239	17-08-2022	319	12-01-2023	356
20-07-2021	209	30-05-2022	652	14-05-2023	514
06-01-2021	130	18-08-2022	220	03-06-2023	255
15-03-2021	283	07-12-2022	291	30-09-2023	317
14-01-2021	260	20-04-2022	216	13-06-2023	606
11-07-2021	133	25-10-2022	504	07-04-2023	534
23-10-2021	159	01-05-2022	240	08-02-2023	112
24-09-2021	478	17-04-2022	711	20-05-2023	631
06-07-2021	204	22-12-2022	155	19-07-2023	297
22-04-2021	435	14-01-2022	80	25-10-2023	286
24-10-2021	501	17-06-2022	202	01-02-2023	120
17-04-2021	678	10-05-2022	402	18-09-2023	213
18-01-2021	102	22-07-2022	293	12-04-2023	996
21-12-2021	155	07-03-2022	587	05-06-2023	279
12-09-2021	174	15-04-2022	526	03-05-2023	1057
21-12-2021	97	27-03-2022	1102	06-02-2023	87
08-06-2021	495	31-07-2022	468	26-08-2023	77
04-04-2021	244	03-05-2022	275	09-02-2023	109
04-01-2021	112	01-02-2022	400	01-03-2023	127
05-03-2021	600	22-12-2022	270	11-06-2023	305
12-05-2021	379	30-05-2022	734	12-02-2023	387
02-05-2021	369	27-05-2022	247	20-05-2023	486
13-09-2021	207	09-06-2022	141	22-09-2023	193
23-06-2021	703	10-06-2022	292	14-11-2023	228
07-12-2021	396	07-05-2022	397	21-06-2023	275
01-08-2021	103	26-04-2022	379	08-08-2023	196
15-04-2021	514	01-09-2022	190	29-03-2023	669
18-06-2021	583	14-05-2022	588	18-07-2023	137
25-11-2021	112	25-09-2022	363	23-05-2023	645
12-07-2021	272	15-05-2022	244	29-08-2023	672
20-01-2021	260	13-06-2022	535	14-12-2023	100

08-05-2021	750	11-10-2022	430	24-06-2023	210
25-10-2021	195	07-09-2022	146	01-10-2023	114
16-07-2021	219	09-06-2022	459	05-08-2023	206
23-03-2021	474	12-10-2022	776	03-11-2023	207
24-10-2021	237	31-10-2022	266	25-07-2023	189
03-03-2021	528	06-06-2022	598	22-06-2023	237
18-06-2021	547	22-07-2022	137	13-09-2023	520
10-09-2021	240	13-01-2022	199	29-08-2023	306
16-06-2021	135	23-12-2022	339	05-03-2023	685
01-12-2021	283	21-07-2022	137	02-12-2023	412
27-11-2021	305	25-03-2022	811	30-10-2023	111
06-03-2021	191	01-05-2022	161	11-12-2023	232
03-04-2021	714	11-06-2022	421	16-04-2023	658
28-06-2021	444	11-03-2022	649	12-03-2023	882
13-05-2021	992	14-03-2022	203	09-08-2023	173
17-12-2021	188	23-04-2022	597	21-07-2023	285
10-12-2021	217	13-01-2022	294	08-11-2023	84
22-12-2021	125	17-08-2022	133	18-02-2023	145
13-11-2021	80	20-05-2022	628	25-10-2023	264
03-01-2021	120	31-10-2022	390	30-05-2023	522
18-10-2021	568	11-10-2022	482	11-04-2023	1039
29-03-2021	702	06-12-2022	139	27-07-2023	182
04-06-2021	366	06-09-2022	260	29-06-2023	675
14-01-2021	180	07-12-2022	138	30-09-2023	454
09-08-2021	727	22-06-2022	643	01-01-2023	220
13-06-2021	388	30-09-2022	216	31-05-2023	433
12-11-2021	171	11-10-2022	294	22-11-2023	562
26-03-2021	399	28-04-2022	315	10-01-2023	230
13-09-2021	584	19-11-2022	301	03-06-2023	166
22-03-2021	323	17-02-2022	145	30-04-2023	315
17-11-2021	458	04-01-2022	172	03-10-2023	514
20-06-2021	749	06-08-2022	137	14-04-2023	396
09-02-2021	246	19-12-2022	138	31-01-2023	355
22-01-2021	250	11-11-2022	140	08-01-2023	188
10-07-2021	273	05-05-2022	390	20-04-2023	259
22-11-2021	63	26-03-2022	803	11-06-2023	328
08-11-2021	217	13-04-2022	1026	29-01-2023	185
16-11-2021	166	29-11-2022	242	16-03-2023	429
16-05-2021	424	24-03-2022	220	29-01-2023	367
26-10-2021	356	27-07-2022	519	08-02-2023	369
02-12-2021	356	20-03-2022	641	22-07-2023	301
14-10-2021	171	01-03-2022	328	08-05-2023	127
12-09-2021	283	18-11-2022	352	10-07-2023	231
14-04-2021	165	22-02-2022	121	20-07-2023	314
24-08-2021	374	04-03-2022	406	25-07-2023	500
08-10-2021	73	27-06-2022	606	22-10-2023	364
04-12-2021	107	16-04-2022	475	12-11-2023	221
10-11-2021	224	19-05-2022	376	21-07-2023	254
13-10-2021	230	20-12-2022	399	05-11-2023	237

03-01-2021	594	10-06-2022	560	01-07-2023	316
14-12-2021	110	27-06-2022	670	13-03-2023	594
07-02-2021	383	20-07-2022	339	17-01-2023	371
12-03-2021	190	20-06-2022	685	03-08-2023	496
03-11-2021	342	23-07-2022	381	25-08-2023	432
26-09-2021	664	25-05-2022	238	09-07-2023	188

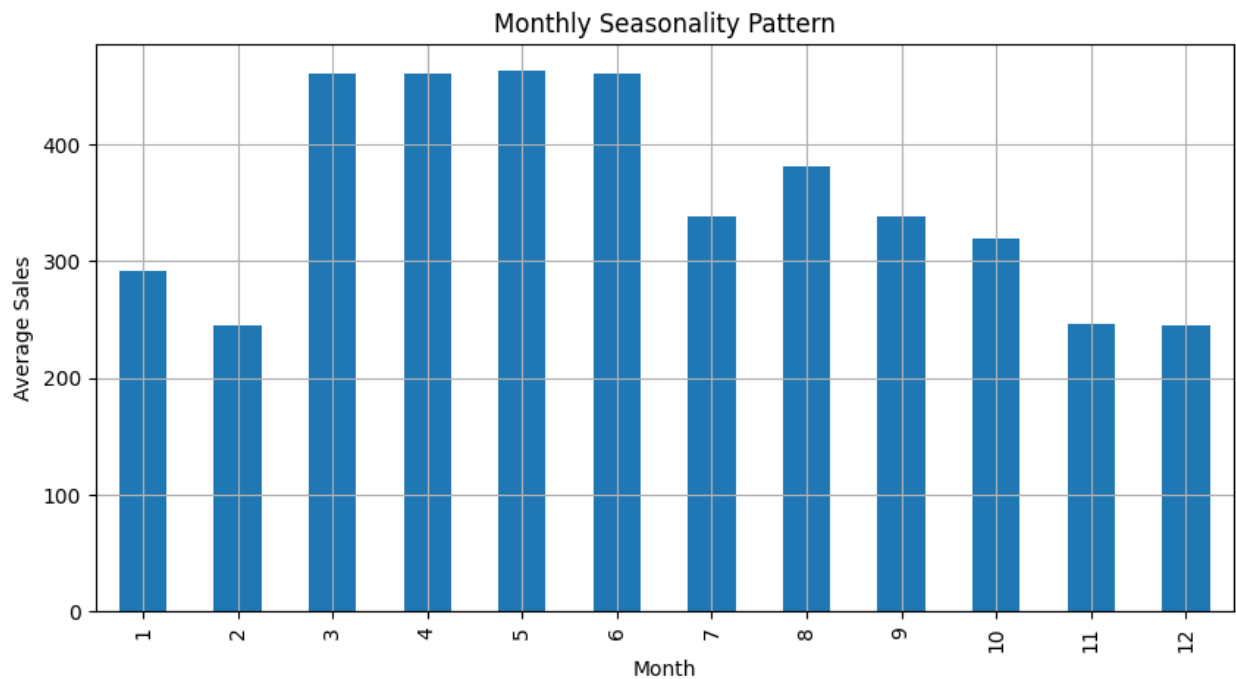
2024 – 2025 Datasets

Date	Sales Qty	Date	Sales Qty
22-08-2024	381	13-05-2025	844
29-02-2024	220	07-09-2025	312
27-06-2024	292	31-10-2025	658
25-04-2024	578	14-08-2025	172
29-12-2024	212	05-09-2025	388
23-05-2024	369	25-04-2025	241
06-08-2024	281	09-10-2025	356
03-06-2024	618	15-12-2025	296
23-06-2024	873	17-10-2025	195
10-03-2024	436	30-08-2025	254
12-07-2024	684	14-05-2025	562
19-06-2024	564	08-09-2025	225
09-05-2024	537	18-03-2025	464
30-01-2024	116	12-12-2025	128
05-08-2024	113	26-09-2025	267
06-04-2024	617	26-07-2025	520
14-02-2024	384	15-11-2025	148
05-01-2024	119	28-11-2025	101
06-01-2024	532	16-05-2025	709
25-11-2024	436	27-08-2025	171
31-08-2024	237	04-02-2025	362
09-10-2024	172	12-02-2025	230
15-11-2024	342	20-04-2025	311
30-04-2024	436	17-06-2025	541
19-08-2024	536	22-10-2025	79
19-04-2024	132	08-01-2025	418
09-02-2024	191	14-04-2025	364
30-09-2024	219	21-05-2025	261
19-04-2024	207	05-10-2025	302
15-01-2024	440	25-11-2025	113
03-02-2024	273	03-05-2025	112
28-07-2024	537	05-03-2025	642
26-10-2024	339	08-05-2025	258
22-10-2024	527	20-01-2025	200
18-02-2024	383	01-11-2025	184
25-04-2024	100	24-08-2025	748
24-02-2024	116	23-04-2025	837
09-02-2024	342	27-11-2025	222

25-04-2024	670	17-03-2025	1098
11-01-2024	97	05-02-2025	172
04-05-2024	207	05-08-2025	358
26-03-2024	790	07-11-2025	222
06-09-2024	100	29-10-2025	296
18-10-2024	177	12-01-2025	135
30-07-2024	351	09-08-2025	313
04-08-2024	75	14-12-2025	152
30-09-2024	442	18-11-2025	171
20-07-2024	163	24-02-2025	128
17-04-2024	276	18-08-2025	101
22-01-2024	51	18-02-2025	117
07-09-2024	409	27-01-2025	448
10-12-2024	111	28-12-2025	529
29-03-2024	865	09-09-2025	284
19-05-2024	181	19-03-2025	316
30-01-2024	603	02-09-2025	81
12-03-2024	532	03-08-2025	178
23-07-2024	165	28-12-2025	148
13-01-2024	507	27-04-2025	355
14-02-2024	373	05-03-2025	522
02-02-2024	350	25-09-2025	106
03-07-2024	338	18-01-2025	212
22-05-2024	442	12-04-2025	505
10-07-2024	270	20-09-2025	285
10-01-2024	239	02-12-2025	360
10-11-2024	386	07-12-2025	138
14-06-2024	286	16-05-2025	255
04-09-2024	530	08-03-2025	367
28-10-2024	308	21-11-2025	229
15-09-2024	224	28-01-2025	223
24-11-2024	453	21-03-2025	336
19-10-2024	505	11-04-2025	442
07-01-2024	538	06-02-2025	370
25-06-2024	430	10-10-2025	416
29-03-2024	630	10-11-2025	241
30-04-2024	685	08-12-2025	530
18-03-2024	198	11-12-2025	425
23-09-2024	168	06-08-2025	168
03-03-2024	438	05-09-2025	389
09-10-2024	486	05-03-2025	334
22-09-2024	372	03-03-2025	822
13-12-2024	150	20-05-2025	567
20-12-2024	580	27-10-2025	132
04-02-2024	266	20-05-2025	318
03-01-2024	129	12-06-2025	882
11-02-2024	165	01-11-2025	92
05-02-2024	126	27-06-2025	372
26-06-2024	843	08-10-2025	273

18-02-2024	188	07-12-2025	82
22-02-2024	112	18-05-2025	693
25-10-2024	225	19-07-2025	508
10-01-2024	163	26-08-2025	765
25-04-2024	469	10-02-2025	323
04-11-2024	392	07-02-2025	176
18-05-2024	609	29-03-2025	670
22-03-2024	761	21-08-2025	271
29-06-2024	748	19-01-2025	279
16-09-2024	497	23-07-2025	219
10-07-2024	376	01-06-2025	670
31-05-2024	417	21-05-2025	307

Source: Podaran Foods India Private Limited



Interpretation:

The graph shows monthly average sales, highlighting seasonal demand patterns. Higher sales during certain months indicate peak consumption periods, especially during summer.

4.6.5 Marketing Spend vs Sales

Table no 4.5.6 (2021 – 2023 Datasets)

Marketing_Spend	Sales_Qty	Marketing_Spend	Sales_Qty	Marketing_Spend	Sales_Qty
2512	333	3455	370	2612	450
3311	495	2894	547	4679	220
4707	267	2503	278	3296	110
1751	246	1224	777	2235	393
3355	262	3081	669	3603	108
7326	199	4824	579	4291.5	481
7380	675	3721	818	1737	801
2735	115	3217	564	5005.5	240
2680	282	2843	342	2761	168
4416	516	1247	312	2589	233
3227	202	4854	383	1527	577
2018	330	2425.5	818	3396	681
3877	222	4282	407	6952.5	159
5872.5	239	2970	319	2646	356
3489	209	6664.5	652	1743	514
4595	130	1070	220	1606.5	255
6625.5	283	1178	291	3936	317
4593	260	6960	216	7281	606
3331	133	2520	504	4518	534
1797	159	2122.5	240	3516	112
2123	478	1665	711	4146	631
2611	204	4017	155	1594	297
1519.5	435	2534	80	2136	286
2721	501	5055	202	4649	120
2392.5	678	6310.5	402	4074	213
3150	102	3588	293	1588.5	996
2422	155	3984	587	5710.5	279
4864	174	3540	526	2164.5	1057
2627	97	7083	1102	2154	87
4269	495	2215	468	2298	77
3703.5	244	6915	275	1556	109
3530	112	1196	400	2301	127
6219	600	1809	270	1743	305
4692	379	5850	734	4946	387
2959.5	369	6883.5	247	6922.5	486
2802	207	4618.5	141	2477	193
2440.5	703	6048	292	2340	228
3087	396	6573	397	7140	275
2591	103	4873.5	379	1845	196
5484	514	1933	190	5454	669
3759	583	5920.5	588	4909	137
1298	112	2090	363	5833.5	645
4736	272	3165	244	3376	672
3975	260	5379	535	2051	100

4377	750	2711	430	5085	210
4709	195	4537	146	3657	114
3158	219	6324	459	1603	206
4527	474	3899	776	2526	207
1301	237	4336	266	4314	189
2839.5	528	2058	598	3360	237
5991	547	4293	137	3016	520
4127	240	4959	199	4521	306
2535	135	1233	339	5989.5	685
4929	283	2223	137	1624	412
4499	305	4468.5	811	4690	111
2947.5	191	6286.5	161	2043	232
4042.5	714	1795.5	421	1641	658
5293.5	444	5236.5	649	3582	882
1603.5	992	3708	203	4356	173
2202	188	7470	597	3584	285
2318	217	1518	294	3063	84
4568	125	1188	133	2684	145
4884	80	1593	628	4568	264
3832	120	3776	390	4356	522
2347	568	1458	482	2157	1039
7497	702	4426	139	1773	182
6330	366	2723	260	3930	675
3750	180	2870	138	1320	454
4407	727	2665.5	643	3850	220
4543.5	388	4700	216	2194.5	433
1539	171	3574	294	3618	562
2526	399	5304	315	2145	230
4710	584	4281	301	3144	166
7287	323	3837	145	7492.5	315
2322	458	1560	172	1322	514
5140.5	749	1987	137	3852	396
3855	246	3570	138	1413	355
4343	250	3451	140	1033	188
4588	273	6487.5	390	3462	259
4716	63	6940.5	803	4530	328
4888	217	4380	1026	1958	185
4472	166	2447	242	6378	429
6160.5	424	2244	220	4876	367
3361	356	4329	519	2581	369
2586	356	3085.5	641	4594	301
1652	171	1696.5	328	4831.5	127
4479	283	3358	352	4551	231
6078	165	1326	121	4394	314
4829	374	3409.5	406	1413	500
3410	73	4824	606	3106	364
4339	107	2566.5	475	4639	221
2304	224	7096.5	376	2191	254
3986	230	4491	399	3218	237

3212	594	4864.5	560	3731	316
4964	110	6595.5	670	5040	594
4956	383	2645	339	4160	371
4336.5	190	4938	685	3254	496
4303	342	4505	381	1386	432
3146	664	2509.5	238	4415	188

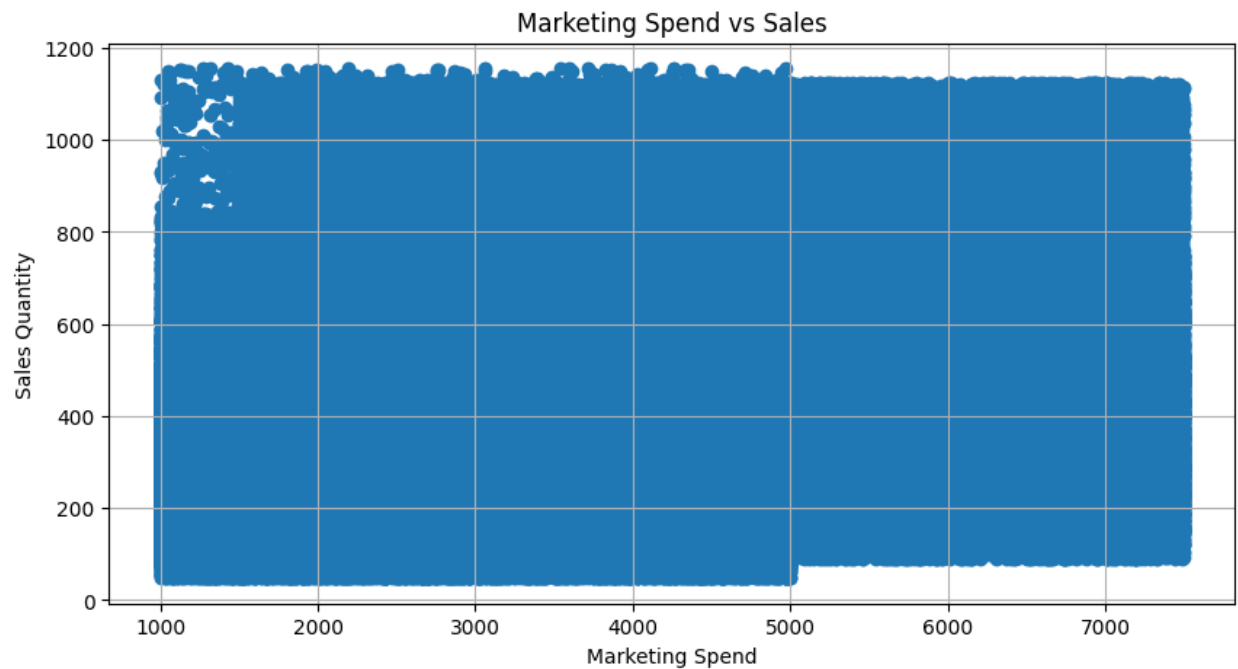
2024 – 2025 Datasets

Marketing Spend	Sales Qty	Marketing Spend	Sales Qty
4051	381	6870	844
2849	220	1905	312
6706.5	292	1666	658
5815.5	578	4438	172
4733	212	4432	388
3001.5	369	2175	241
2400	281	2685	356
2755.5	618	4353	296
7275	873	3527	195
5112	436	3401	254
1565	684	6714	562
5950.5	564	3676	225
5319	537	1824	464
1194	116	4140	128
3180	113	2772	267
6723	617	1888	520
3032	384	3143	148
4842	119	1571	101
1856	532	1540.5	709
2853	436	3583	171
1390	237	4195	362
2932	172	4647	230
1280	342	6871.5	311
3030	436	5341.5	541
4489	536	3259	79
4785	132	3045	418
2689	191	6402	364
4904	219	5616	261
2620.5	207	1646	302
2496	440	3203	113
1303	273	6309	112
2223	537	6822	642
2976	339	3483	258
4975	527	4185	200
2376	383	1912	184
7182	100	3257	748
4125	116	6055.5	837
4304	342	4758	222

5782.5	670	3309	1098
2343	97	2995	172
5392.5	207	2936	358
1951.5	790	1401	222
4514	100	2228	296
2247	177	4366	135
2870	351	2620	313
1802	75	3208	152
2504	442	4054	171
3398	163	3950	128
1944	276	3767	101
2500	51	3115	117
1050	409	1709	448
1824	111	3117	529
6226.5	865	2374	284
1813.5	181	7488	316
1341	603	3675	81
4243.5	532	2111	178
3205	165	1904	148
1776	507	3637.5	355
1739	373	4377	522
2009	350	4803	106
3285	338	1099	212
5082	442	6279	505
3256	270	1434	285
1001	239	1594	360
1283	386	4697	138
2446.5	286	4840.5	255
4716	530	5580	367
4547	308	1821	229
4254	224	1058	223
4210	453	5626.5	336
3417	505	7129.5	442
2679	538	3325	370
5001	430	4375	416
3103.5	630	2941	241
5980.5	685	2460	530
3592.5	198	2199	425
4983	168	4941	168
3841.5	438	3741	389
2433	486	2437.5	334
4884	372	7033.5	822
1025	150	5044.5	567
2931	580	1079	132
1500	266	5418	318
1962	129	6856.5	882
3155	165	3062	92
3922	126	4903.5	372
2490	843	4890	273

2779	188	4641	82
4133	112	4009.5	693
4113	225	2388	508
3823	163	2364	765
6039	469	2835	323
1209	392	4062	176
6813	609	6145.5	670
5310	761	3478	271
4639.5	748	2337	279
1227	497	1999	219
4030	376	1809	670
4219.5	417	6994.5	307

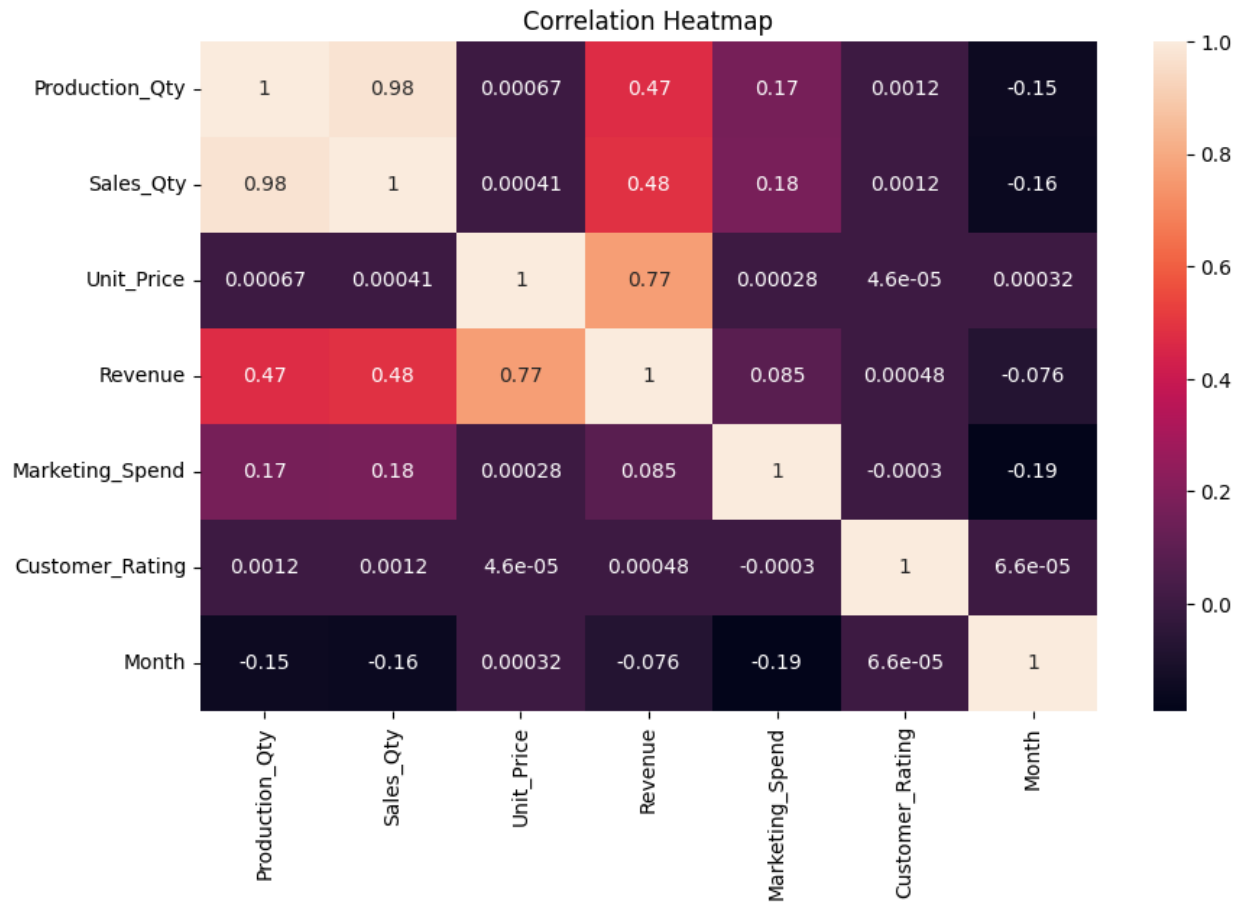
Source: Podaran Foods India Private Limited



Interpretation:

The graph shows the relationship between marketing expenditure and sales. It indicates that higher marketing efforts generally lead to increased sales.

4.6.6 Correlation Heatmap



Interpretation:

The heatmap shows the relationship between variables. Strong correlations indicate important factors influencing sales and demand. The highest correlation heatmap is 0.98 for the relationship between the Production quantity and sales quantity then the lowest correlation heatmap is -0.15 for the Month and production quantity.

4.7 Forecasting Analysis

Table no 4.5.7 (2021 – 2023 Datasets)

Date	Sales_Qty	Date	Sales_Qty	Date	Sales_Qty
02-11-2021	333	14-11-2022	370	19-10-2023	450
05-11-2021	495	16-02-2022	547	30-10-2023	220
10-03-2021	267	27-01-2022	278	06-02-2023	110
03-09-2021	246	02-01-2022	777	23-04-2023	393
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2024 – 2025 Datasets

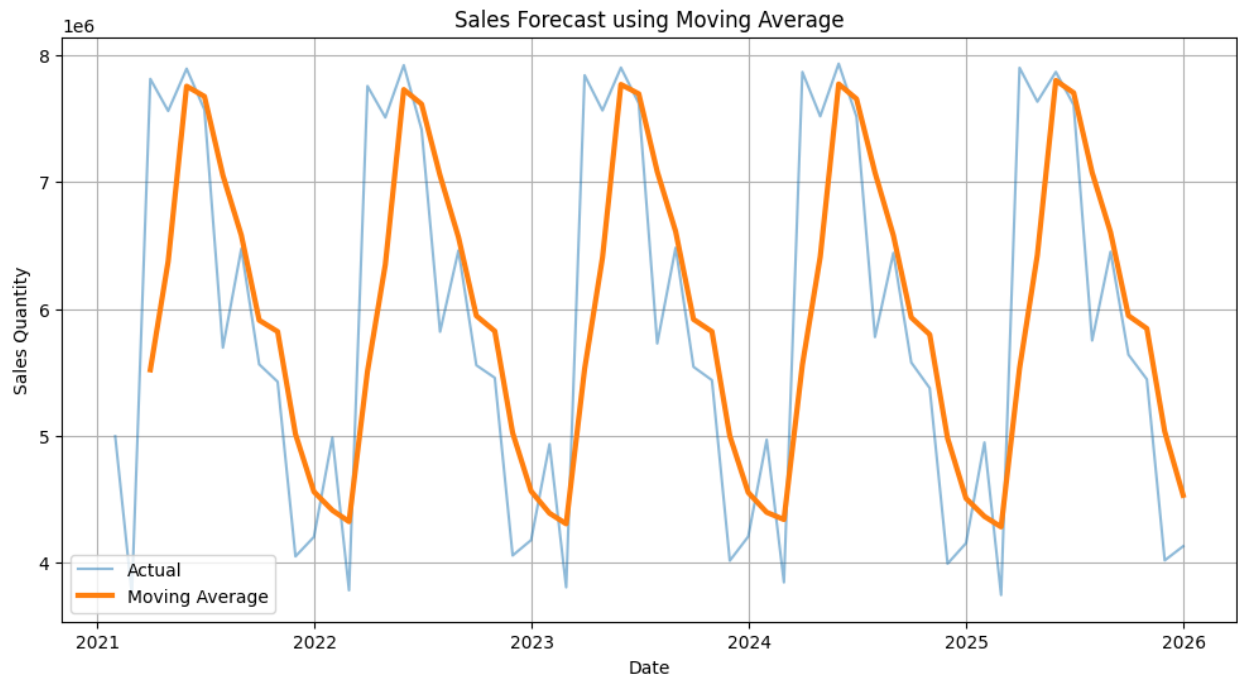
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09-05-2024	537	18-03-2025	464
30-01-2024	116	12-12-2025	128
05-08-2024	113	26-09-2025	267
06-04-2024	617	26-07-2025	520
14-02-2024	384	15-11-2025	148
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28-07-2024	537	05-03-2025	642
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03-01-2024	129	12-06-2025	882
11-02-2024	165	01-11-2025	92
05-02-2024	126	27-06-2025	372
26-06-2024	843	08-10-2025	273

18-02-2024	188	07-12-2025	82
22-02-2024	112	18-05-2025	693
25-10-2024	225	19-07-2025	508
10-01-2024	163	26-08-2025	765
25-04-2024	469	10-02-2025	323
04-11-2024	392	07-02-2025	176
18-05-2024	609	29-03-2025	670
22-03-2024	761	21-08-2025	271
29-06-2024	748	19-01-2025	279
16-09-2024	497	23-07-2025	219
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Source: Podaran Foods India Private Limited

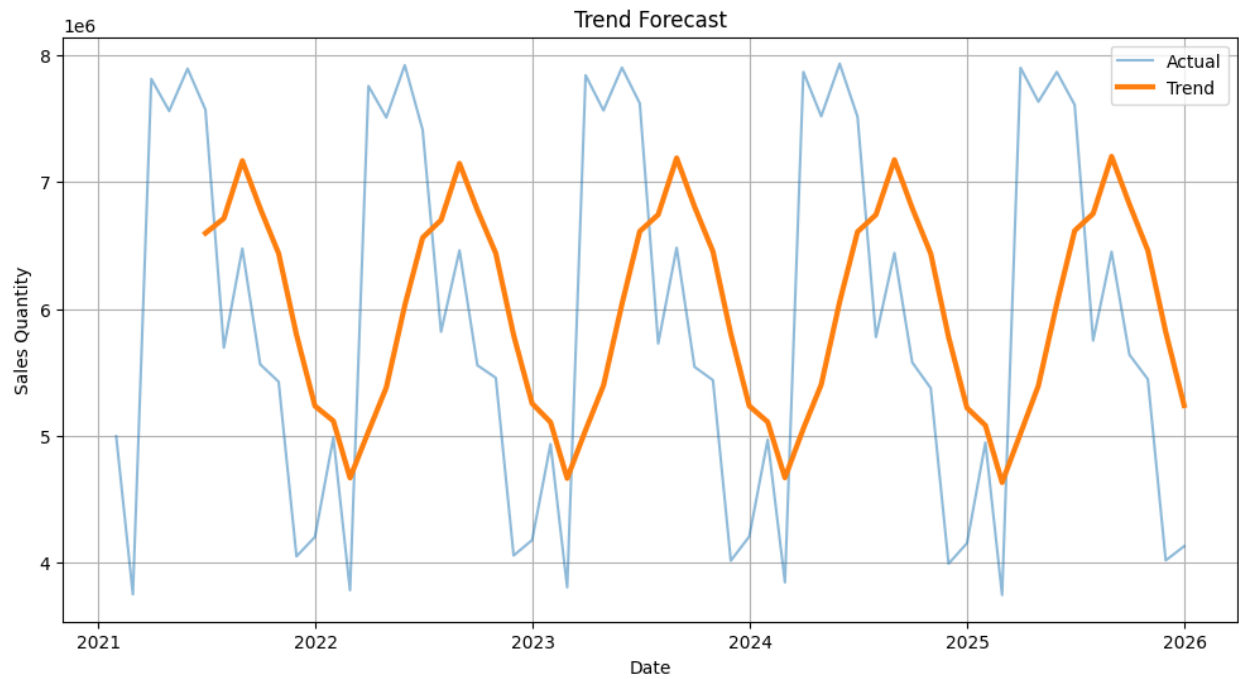
4.7.1 Moving Average Forecast



Interpretation:

The moving average smooths out fluctuations in the data and helps identify trends. It is useful for short-term forecasting of sales. The moving average is remain similar for the all the five years.

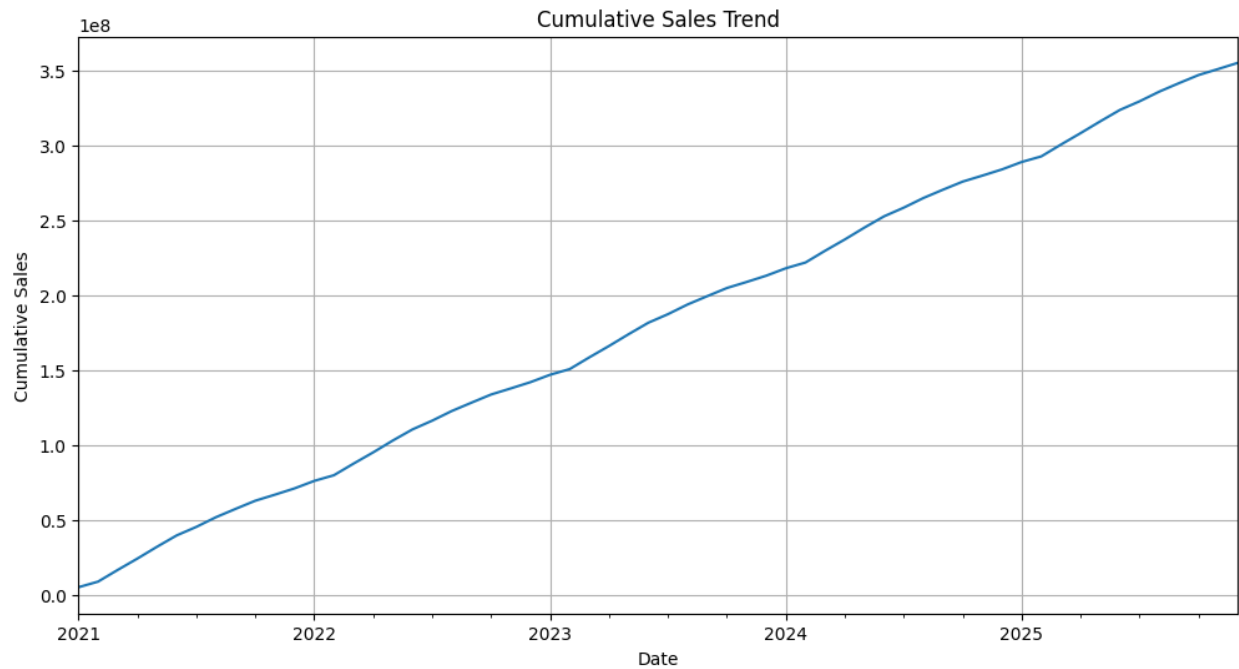
4.7.2 Trend Forecast



Interpretation:

The trend line shows long-term movement in sales. It helps in understanding the direction of future demand and supports strategic planning. This indicates the upcoming years are follow the same trend according to this chart.

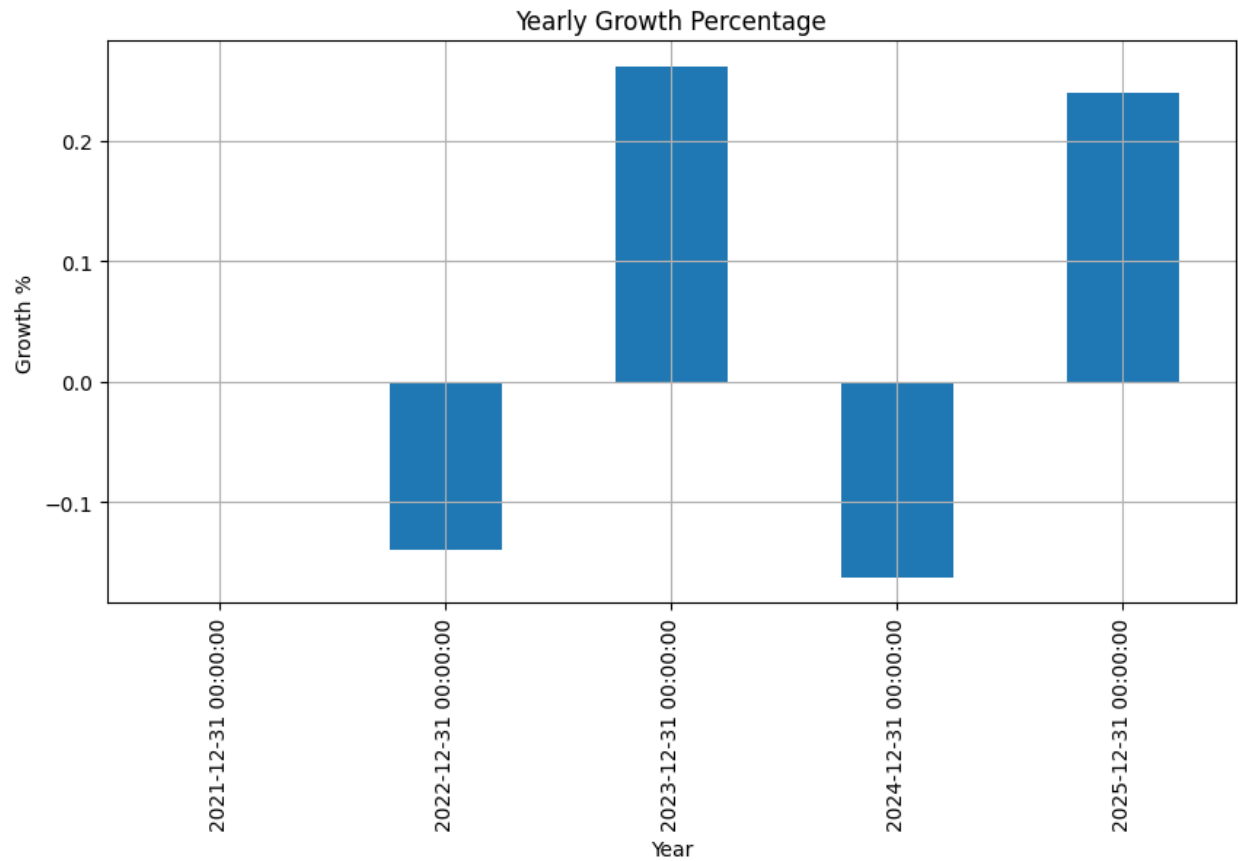
4.7.3 Cumulative Sales Trend



Interpretation:

The cumulative graph shows the total growth in sales over time. It reflects overall business expansion and performance. It shows from the year 2021 to 2025 it represents upward trend in the cumulative sales.

4.7.4 Yearly Growth Analysis



Interpretation:

The graph shows year-wise growth percentage. It helps evaluate the performance of the company and indicates periods of growth and decline. The reason for there is no chart representation for 2021 is because of the there is comparable previous year data so the chart begins with the year 2022 and it followed till 2025.

CHAPTER V

FINDINGS, SUGGESTIONS AND CONCLUSION

5.1 FINDINGS

5.1.1. Sales Analysis

- **Weekly Sales Trend**

The weekly sales analysis reveals noticeable short-term fluctuations in demand. These variations indicate that customer purchasing behaviour changes frequently across different weeks. The fluctuations may be influenced by factors such as market demand, supply conditions, and promotional activities. This suggests that the company needs flexible planning to manage short-term demand variations effectively.

- **Monthly Sales Trend**

The monthly sales analysis clearly shows strong seasonal patterns in demand. Sales are significantly higher during summer months due to increased consumption of beverage products, while lower sales are observed during off-season periods. This indicates that demand is highly influenced by seasonal factors. The finding highlights the importance of seasonal forecasting and production planning to meet peak demand.

- **Yearly Sales Trend**

The yearly sales trend indicates a gradual increase in sales over the study period. Although minor fluctuations are present, the overall pattern shows consistent growth in business performance. This suggests stable demand and effective market presence. The finding indicates that the company is maintaining steady progress over the years.

5.1.2. Business Analysis

- **Region-wise Sales**

The region-wise analysis shows that Tamil Nadu and Kerala contribute the highest share of total sales. This indicates strong market penetration and higher demand in these regions. Other regions show comparatively lower sales, suggesting potential opportunities for market expansion. This finding helps in focusing marketing and distribution strategies region-wise.

- **Product-wise Sales**

The product-wise analysis reveals that Mango and Cola are the highest-selling products among all variants. This indicates strong customer preference for these products. Other products show relatively lower demand, suggesting the need for improvement in marketing or product positioning. This finding supports decision-making in production planning and product strategy.

- **Bottle Size Distribution**

The bottle size analysis indicates that 200 ml bottles contribute the highest share of total sales. This suggests that customers prefer smaller pack sizes due to affordability and convenience. Larger bottle sizes have lower sales, indicating limited demand. This finding helps in optimizing packaging and production strategies based on customer preference.

- **Seasonal Analysis**

The seasonal analysis confirms that sales are highest during summer months, indicating peak consumption periods for beverage products. The increase in demand during these months reflects the influence of weather conditions on customer purchasing behavior. Lower sales during other months indicate off-season demand. This finding highlights the importance of seasonal planning in production and inventory management.

5.1.3. Marketing and Demand Relationship

- **Marketing Spend vs Sales**

The analysis of marketing spend versus sales shows a weak to moderate positive relationship between the two variables. Although increased marketing expenditure contributes to higher sales in some cases, the relationship is not very strong, as indicated by the scattered distribution of data points. This suggests that marketing alone does not significantly influence sales performance. Other factors such as seasonality, product preference, and pricing also play a major role. Therefore, marketing should be used along with other strategies to improve sales effectively.

- **Correlation Heatmap**

The correlation heatmap shows the relationship between different variables in the dataset. The highest positive correlation (0.98) is observed between production quantity and sales quantity, indicating a strong direct relationship where higher production supports higher sales. The lowest correlation (-0.15) is observed between month and production quantity, indicating a weak negative relationship. This suggests that production is not strictly dependent on monthly variation. Overall, the heatmap helps in identifying key factors influencing sales and highlights that production is the most significant factor affecting sales performance.

5.1.4. Forecasting Analysis

- **Moving Average Forecast**

The moving average analysis smooths out short-term fluctuations in sales data and highlights the overall trend. The results show that the moving average remains relatively stable across the years, indicating consistency in sales patterns. This suggests that demand does not vary drastically in the short term. The method is useful for short-term forecasting and helps in understanding general sales movement without noise.

- **Trend Forecast**

The trend analysis shows the long-term movement of sales over time. The upward trend indicates that sales are gradually increasing over the years. This suggests positive business growth

and stable market demand. The trend line also indicates that future sales are likely to follow a similar pattern if current conditions remain unchanged. This finding supports long-term planning and strategic decision-making.

- **Cumulative Sales Trend**

The cumulative sales graph shows continuous growth in total sales over the study period from 2021 to 2025. The upward trend indicates consistent business expansion and increasing overall performance. This reflects the company's ability to maintain steady demand over time. The graph highlights long-term growth and provides a clear picture of total sales accumulation.

- **Yearly Growth Analysis**

The yearly growth analysis shows the percentage change in sales across different years. It indicates both positive and negative growth trends, reflecting variations in business performance. The absence of data for 2021 is due to the lack of a previous year for comparison, as growth is calculated based on year-to-year changes. The graph begins from 2022 and continues up to 2025. This analysis helps in evaluating performance trends and identifying periods of growth and decline.

5.2 SUGGESTIONS

- The company should increase production during peak seasons to meet high demand.
- More focus should be given to high-performing products such as Mango and Clear Lemon.
- Marketing efforts should be increased in low-performing regions to expand market reach.
- The company should adopt advanced forecasting techniques for better accuracy.
- Inventory management should be improved to reduce excess stock and wastage.
- Promotional strategies should be strengthened to increase customer engagement.

5.3 CONCLUSION

The study concludes that sales and demand forecasting plays a crucial role in improving business performance. The analysis shows that demand is influenced by seasonal factors, product preference and marketing activities. The use of data science techniques has provided valuable insights into sales patterns and future demand trends.

The company shows stable growth and has strong potential for expansion. With proper planning and implementation of data-driven strategies, Podaran Foods India Private Limited can achieve higher efficiency, improved sales performance and long-term sustainability.

The business analysis revealed that Tamil Nadu and Kerala are the major contributors to total sales, indicating strong regional demand. Product-wise analysis showed that Mango and Cola are the most preferred products, while bottle size analysis indicated that smaller pack sizes are more popular among customers. These insights highlight the importance of focusing on product preference, regional strategies, and packaging decisions.

The relationship analysis between marketing spend and sales showed a weak to moderate positive relationship. This indicates that while marketing contributes to sales growth, it is not the only factor influencing demand. Other factors such as seasonality, product demand, and pricing play a significant role in determining sales performance.

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